Sienna WAV — manufacturing package REV B (2026-10-31): frame + ramp

Frame: everything that was 10 ga is now 11 ga (0.1196″); square tubes 11 ga wall. Ramp: aluminum 5052-H32 0.125″. Units inch. Inside bend radius = t, K = 0.44.

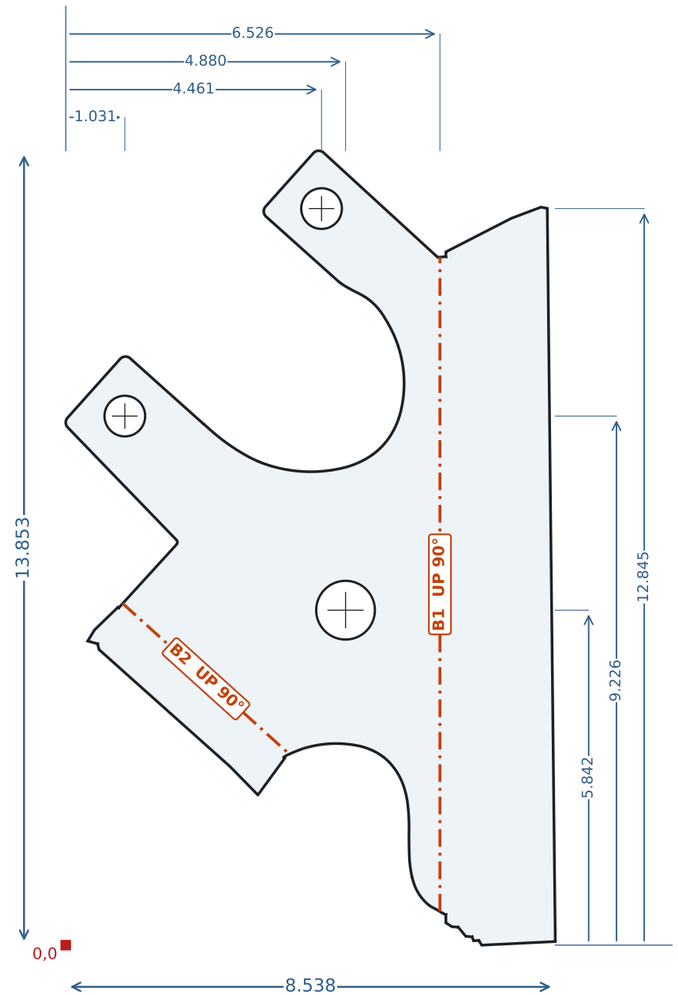
Item	Material / thickness	Flat (in)	Bend lines	Qty / vehicle	lb each	Process
FR-1	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	8.538 × 13.853	2	1 FR + 1 FL	2.04	laser + 2 bends
FR-2	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	22.264 × 3.250	6	1 FR + 1 FL	2.27	laser + 6 bends + weld
FR-3	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	4.084 × 8.000	2	1 FR + 1 FL	1.03	laser + 2 bends
FR-4-1 strip	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	18.827 × 2.746	9	1 FR + 1 FL	1.63	laser + 5 bends (2 radius)
FR-4-2 side A	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	6.500 × 6.875	0	1 FR + 1 FL	0.71	laser
FR-4-3 side B	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	6.500 × 6.875	0	1 FR + 1 FL	0.71	laser
FR-5	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	10.694 × 6.198	2	1 FR + 1 FL	1.41	laser + 2 bends
FR-7	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	9.853 × 8.186	3	1 FR + 1 FL	1.45	laser + 3 bends + weld
FR-8	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	6.209 × 12.000	2	1 FR + 1 FL	1.79	laser + 2 bends
FR-10-1 U-body	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	12.374 × 11.235	3	1 FR + 1 FL	2.29	laser + 3 bends + drill
FR-10-2 strip	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	12.560 × 2.746	3	1 FR + 1 FL	1.05	laser + 3 bends
FR-11	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	5.125 × 2.000	0	2 FR + 2 FL	0.25	laser
FF-1	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	20.142 × 6.870	4	1	3.89	laser + 4 bends + weld
FF-3-1 top	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	39.403 × 15.073	10	1	17.70	laser + 10 bends
FF-3-2 lower	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	39.000 × 14.595	2	1	16.12	laser + 2 bends
FF-3-3 wall	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	4.687 × 1.125	0	2	0.18	laser
FF-3-4 wall	HR steel A1011 CS, 11 ga 0.1196″ (3.04 mm)	5.013 × 1.125	0	2	0.19	laser
RA-1	Aluminum 5052-H32, 0.125″ (3.18 mm)	22.000 × 34.566	2	1	8.07	laser + bends
RA-2	Aluminum 5052-H32, 0.125″ (3.18 mm)	23.533 × 34.941	3	1	9.87	laser + bends
RA-3	Aluminum 5052-H32, 0.125″ (3.18 mm)	19.658 × 45.066	3	1	9.10	laser + bends
RA-4	Aluminum 5052-H32, 0.125″ (3.18 mm)	6.691 × 33.500	2	1	2.69	laser + bends
RA-5	Aluminum 5052-H32, 0.125″ (3.18 mm)	9.783 × 33.500	1	1	2.37	laser + bends
RA-6	Aluminum 5052-H32, 0.125″ (3.18 mm)	3.066 × 30.000	2	1	1.11	laser + bends
RA-7	Aluminum 5052-H32, 0.125″ (3.18 mm)	40.631 × 22.556	4	1	10.30	laser + bends
RA-8	Aluminum 5052-H32, 0.125″ (3.18 mm)	4.625 × 4.066	2	2	0.18	laser + bends
FR-4 tube	DOM OD 1.500 × ID 1.181 (wall 0.160″)	cut 3.689	—	1 FR + 1 FL		saw
FR-6	Sq. tube 3×3 A500, wall 11 ga 0.120″	cut 14.250	—	1 FR + 1 FL		saw
FR-9	Sq. tube 3×3 A500, wall 11 ga 0.120″	cut 8.000	—	1 FR + 1 FL		saw
FR-10 tube	DOM OD 1.500 × ID 1.181 (wall 0.160″)	cut 3.407	—	1 FR + 1 FL		saw
FF-4	Sq. tube 1½×1½ A500, wall 11 ga 0.120″	cut 39.000	—	1		saw

FR-1 · Plate with ears flat pattern (2 bends 90°)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 8.538″ × 13.853″ · ≈ 2.04 lb each · QTY 1 FR + 1 FL = 2 per vehicle

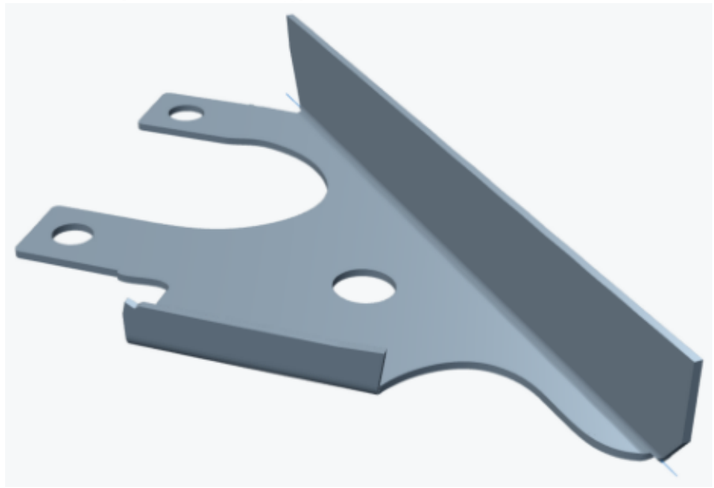
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 6.526
B2	90.0°	UP (toward)	R=t 0.120	2	(0.993, 5.959) (3.900, 3.331)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.709	1.031	9.226	laser
H2	Ø0.709	4.461	12.845	laser
H3	Ø1.024	4.880	5.842	laser

FILES

- FR-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bend 90° up on dashed lines

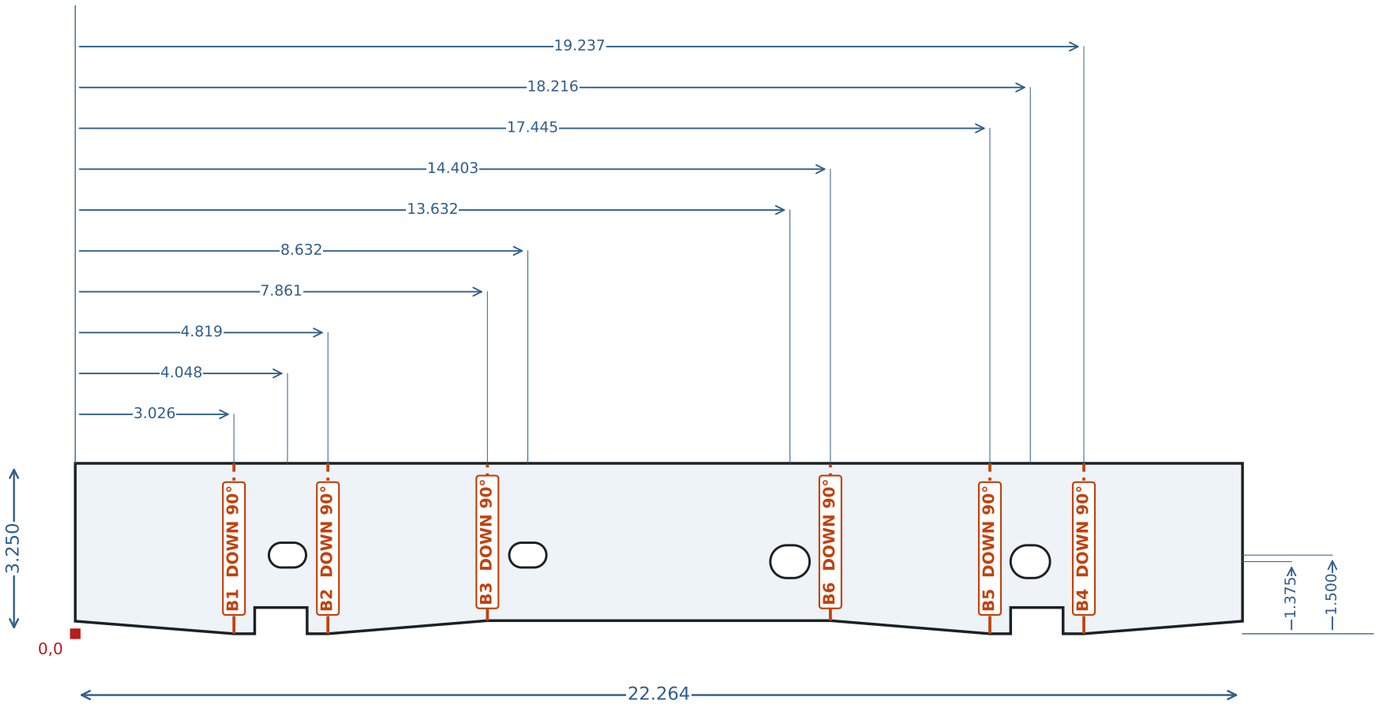
FR-1 Side plate with ears and 2 flanges	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-1 · Plate with ears flat pattern (2 bends 90°)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**

FR-2 · FR-2 one piece (top + 2 U legs)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 22.264″ × 3.250″ · ≈ 2.27 lb each · QTY 1 FR + 1 FL = 2 per vehicle

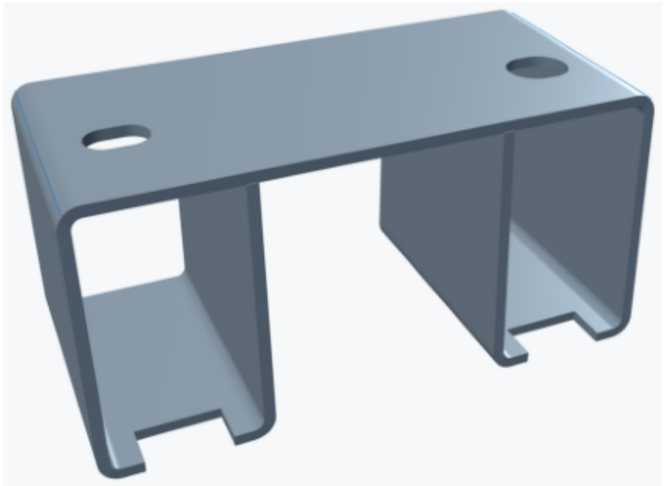
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	1	X = 3.026
B2	90.0°	DOWN (away)	R=t 0.120	3	X = 4.819
B3	90.0°	DOWN (away)	R=t 0.120	5	X = 7.861
B4	90.0°	DOWN (away)	R=t 0.120	2	X = 19.237
B5	90.0°	DOWN (away)	R=t 0.120	4	X = 17.445
B6	90.0°	DOWN (away)	R=t 0.120	6	X = 14.403

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	cut-out 0.708×0.472	4.048	1.500	laser
H2	cut-out 0.709×0.472	8.632	1.500	laser
H3	cut-out 0.750×0.625	13.632	1.375	laser
H4	cut-out 0.750×0.625	18.216	1.375	laser

FILES

- FR-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* one blank, 6 bends 90° the same way (R = t): inner walls first, then the bottoms, then the outer walls; all ovals (top and leg bottoms) laser cut;
½" notch at the front edge of each bottom; inner walls reach the underside of the top (11 ga)

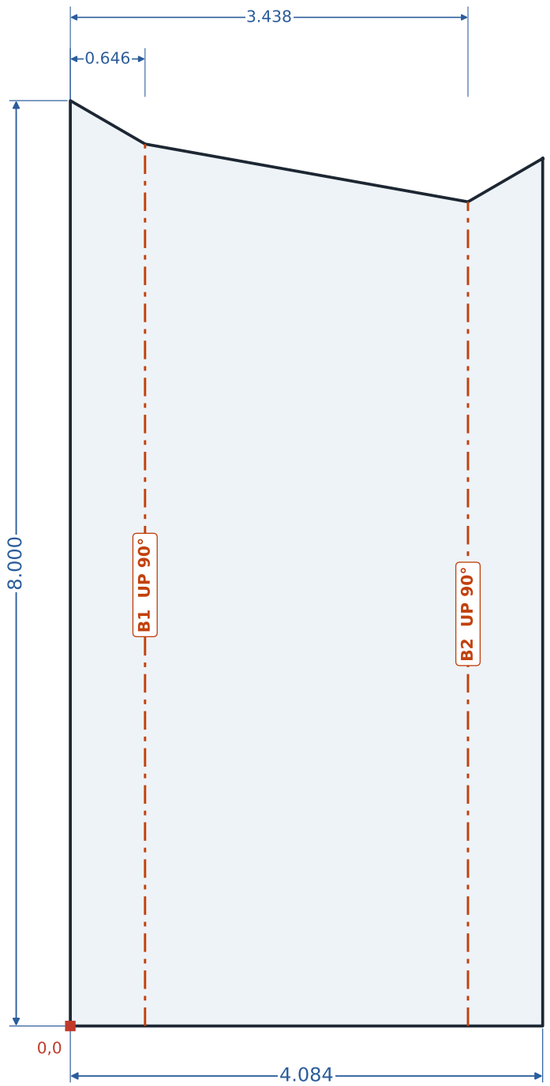
FR-2 Bridge bracket (top + 2 U legs)	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-2 · FR-2 one piece (top + 2 U legs)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**

FR-3 · Channel — flat pattern (2 bends 90°)

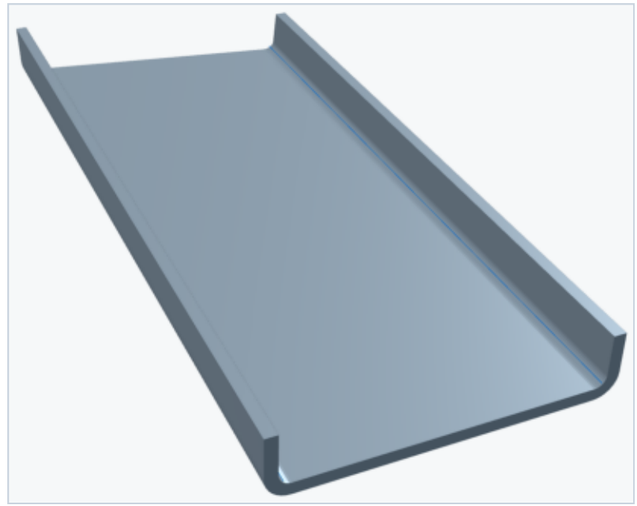
11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.084″ × 8.000″ · ≈ 1.03 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 0.646
B2	90.0°	UP (toward)	R=t 0.120	2	X = 3.438

FILES

- FR-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-3_ref.dxf (reference: + bend lines, drill-later)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bend 90° on dashed lines, 3/4 flanges inward

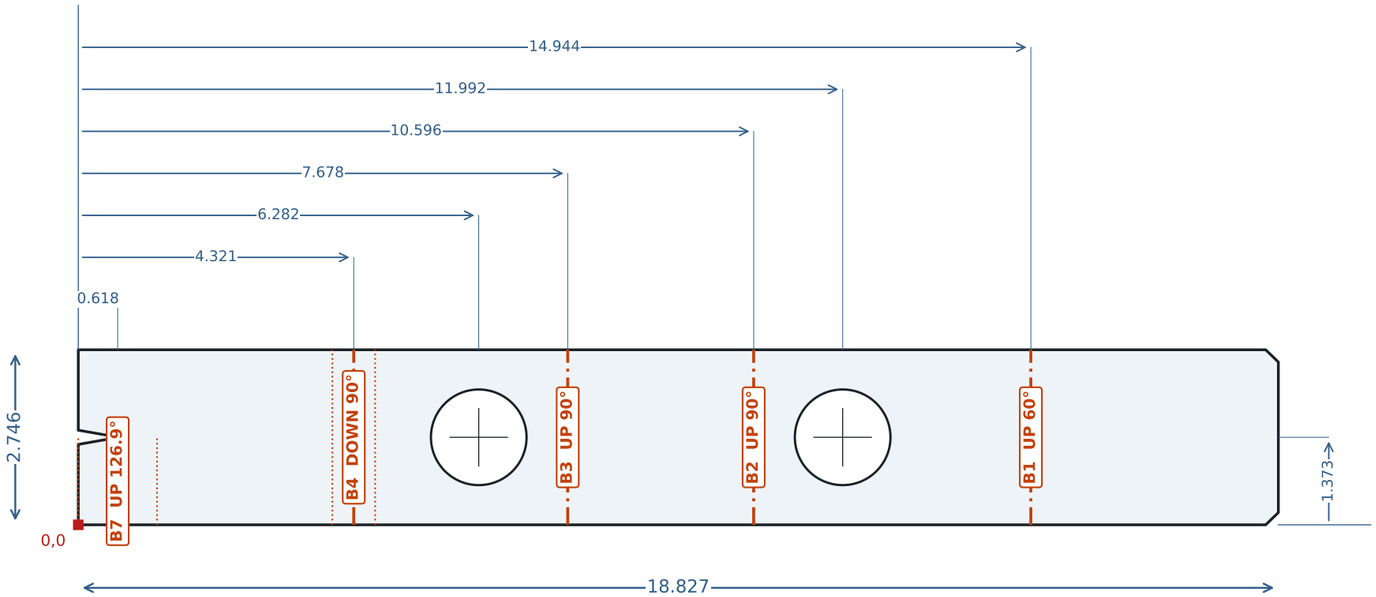
FR-3 Channel	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FR-3 · Channel — flat pattern (2 bends 90°)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/1**

FR-4-1 · Wrap strip (ear, top, front, bottom, slant)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 18.827″ × 2.746″ · ≈ 1.63 lb each · QTY 1 FR + 1 FL = 2 per vehicle

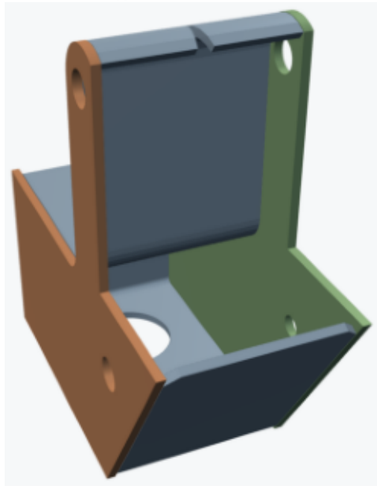
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	60.0°	UP (toward)	R=t 0.120	5	X = 14.944
B2	90.0°	UP (toward)	R=t 0.120	4	X = 10.596
B3	90.0°	UP (toward)	R=t 0.120	3	X = 7.678
B4	90.0°	DOWN (away)	R 0.375	2	X = 4.321
		radius zone	edges		X = 3.986 / 4.657
B7	126.9°	UP (toward)	R 0.505	1	X = 0.618
		radius zone	edges		X = 0.000 / 1.236

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.500	6.282	1.373	laser
H2	Ø1.500	11.992	1.373	laser

FILES

- FR-4-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* arc C-D bent over R 0.505 (outer R 0.625 = side plate arc); R 0.375 at E (concave, opposite way); no corner reliefs; strip width = 3" - 2t - 0.015

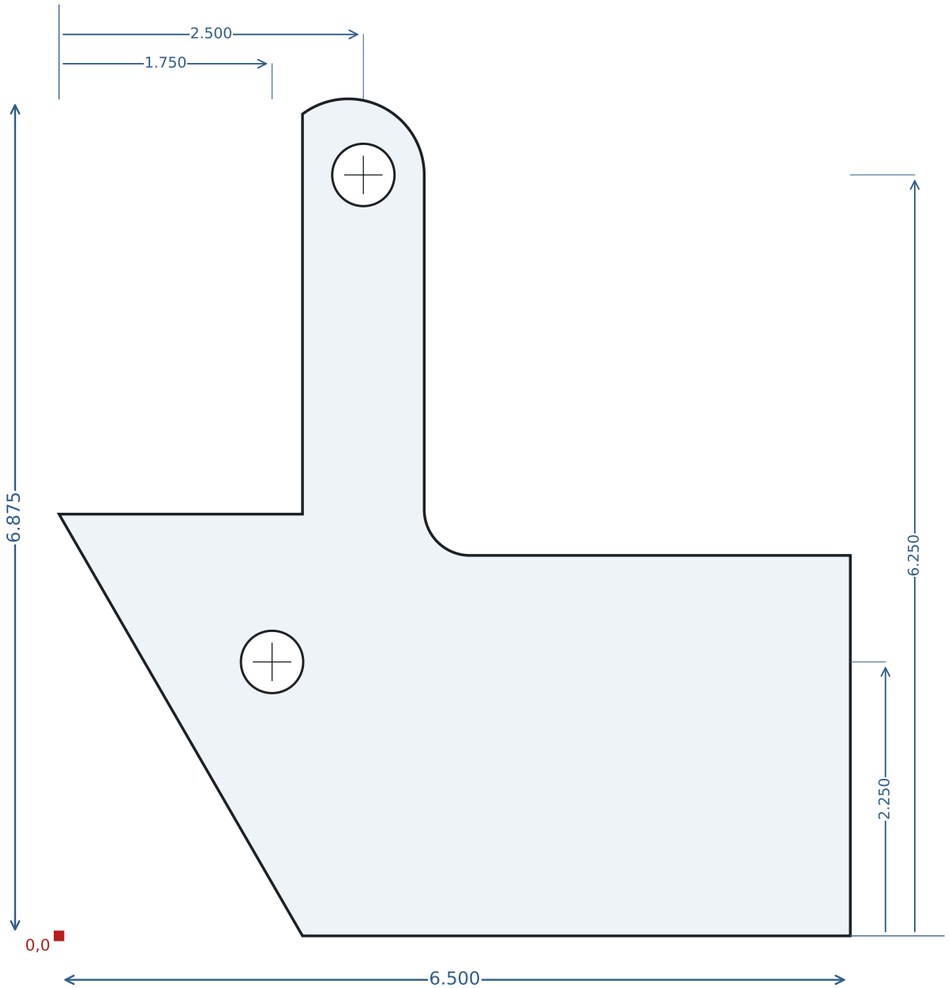
FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-1 · Wrap strip (ear, top, front, bottom, slant)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/4**

FR-4-2 · Side plate A (2 holes Ø0.512)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.500″ × 6.875″ · ≈ 0.71 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)

flat part - no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.512	1.750	2.250	laser
H2	Ø0.512	2.500	6.250	laser

FILES

- FR-4-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* flat, no bends

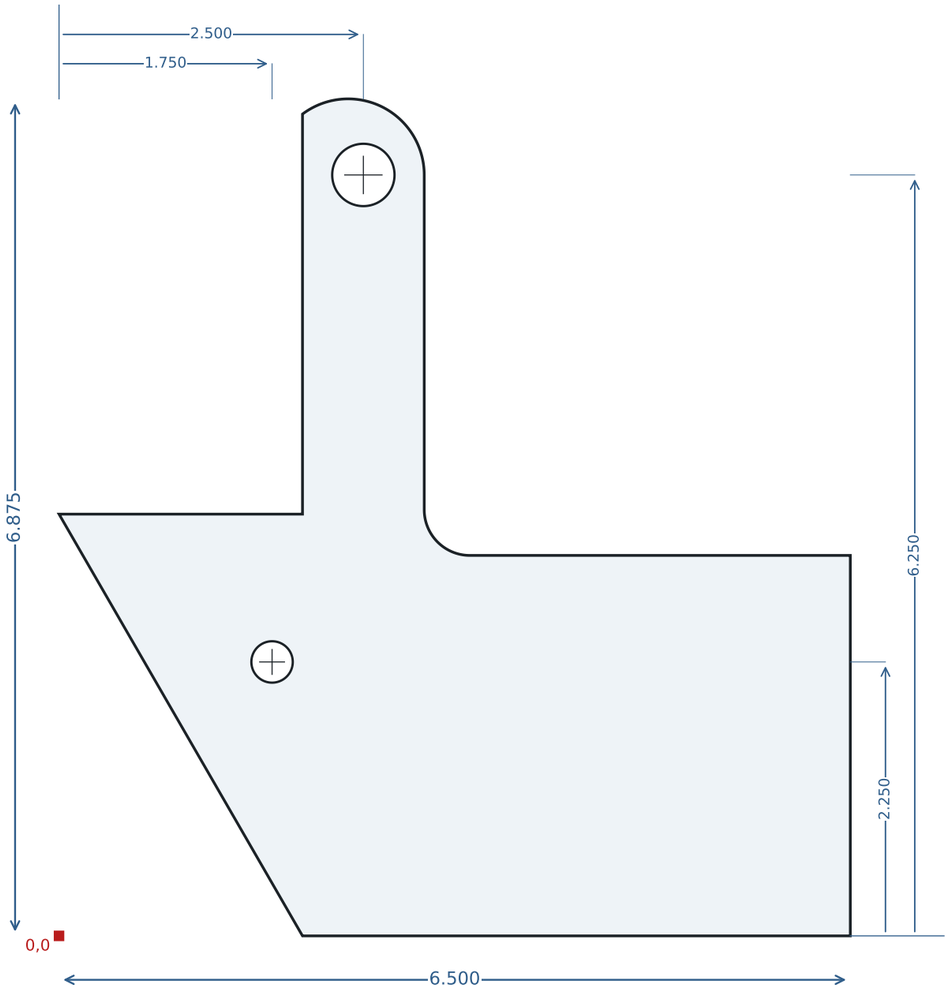
FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-2 · Side plate A (2 holes Ø0.512)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 2/4**

FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.500″ × 6.875″ · ≈ 0.71 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-4-3 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)

flat part - no bends

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.340	1.750	2.250	laser
H2	Ø0.512	2.500	6.250	laser

FILES

- FR-4-3_FR_cut.dxf (right, qty 1 / vehicle)
- FR-4-3_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-4-3_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* flat, no bends

FR-4 Bracket: wrap strip + 2 side plates + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-4-3 · Side plate B (holes Ø0.512 + Ø0.34)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 3/4**

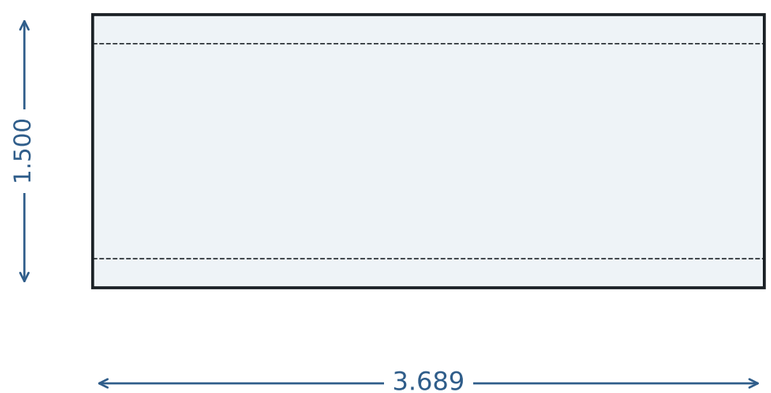
FR-4 · tube cut list

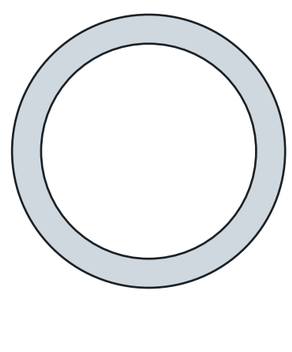
MATERIAL: DOM round steel tube A513

WALL THICKNESS: OD 1.500 × ID 1.181 (wall 0.160″)

Tube OD 1.5 / ID 1.181 cut 3.689″ (through bottom and top)

Stock: DOM round steel tube OD 1.500 × ID 1.181 (wall 0.160″)





section, wall 0.1595″

Cut length 3.689″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

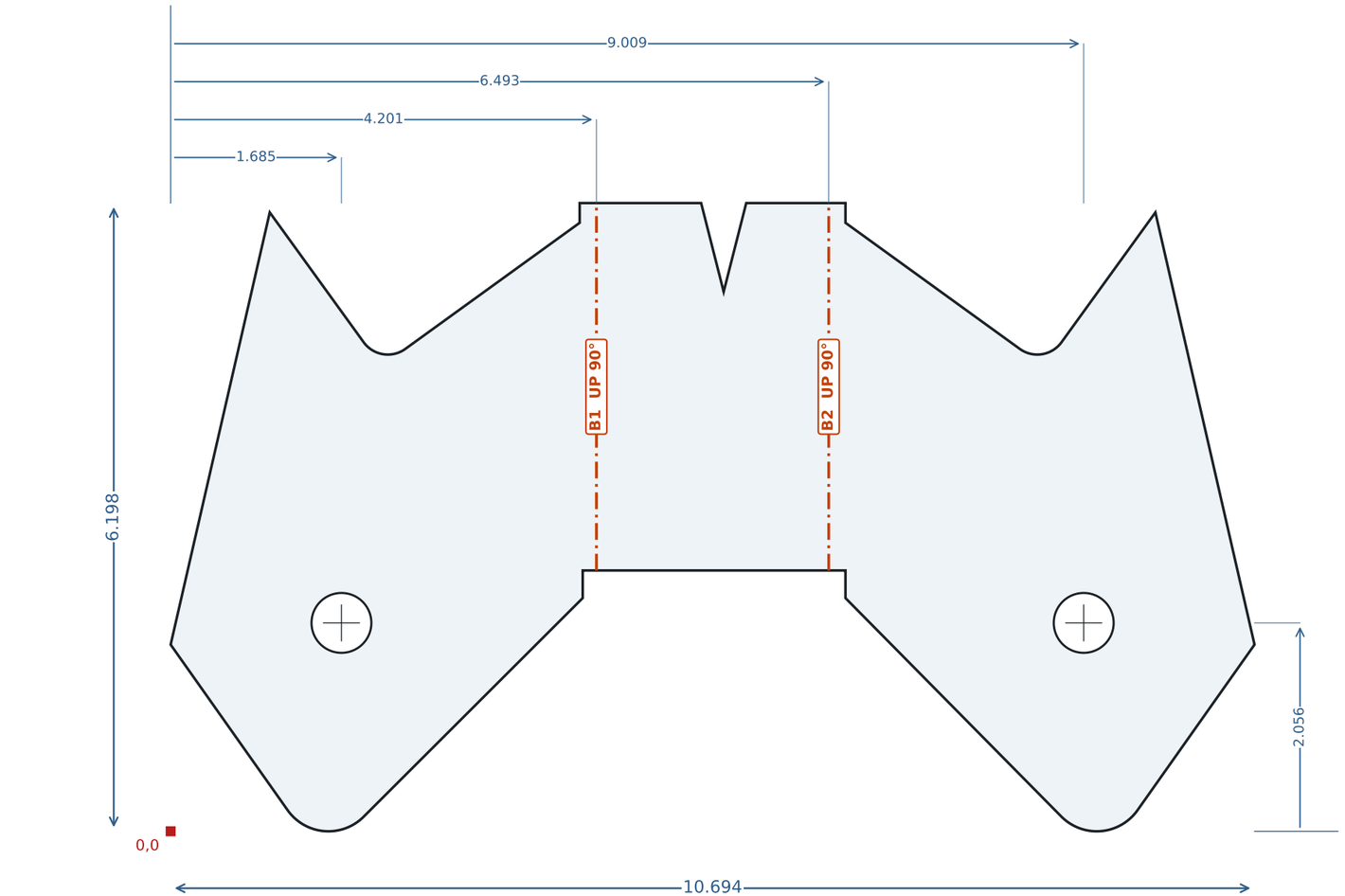
FR-4 Bracket: wrap strip + 2 side plates + tube		**DOM round steel tube A513**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list		**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 4/4**

FR-5 · U-body (2 sides + plate A-C)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 10.694″ × 6.198″ · ≈ 1.41 lb each · QTY 1 FR + 1 FL = 2 per vehicle

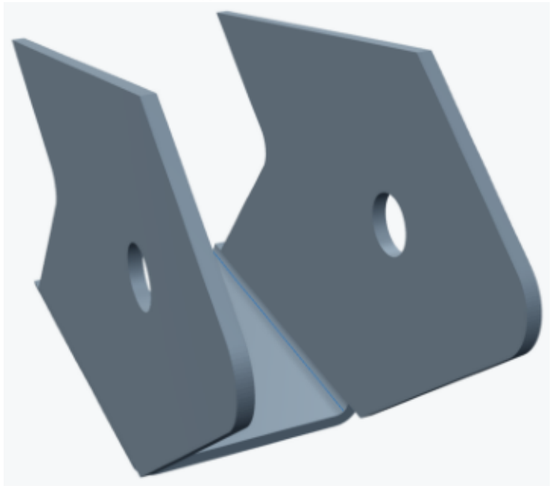
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-5 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 4.201
B2	90.0°	UP (toward)	R=t 0.120	2	X = 6.493

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	1.685	2.056	laser
H2	Ø0.591	9.009	2.056	laser

FILES

- FR-5_FR_cut.dxf (right, qty 1 / vehicle)
- FR-5_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-5_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* 90° bends, R = t

FR-5 U-body bracket	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-5 · U-body (2 sides + plate A-C)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**

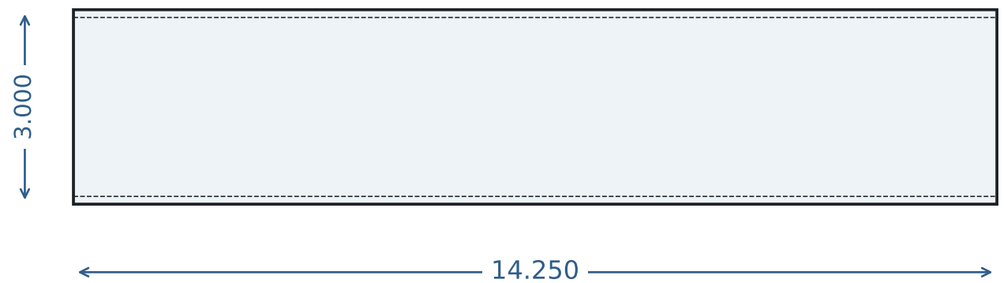
FR-6 · tube cut list

MATERIAL: Square steel tube A500, 3 × 3

WALL THICKNESS: 11 ga 0.120″ (3.04 mm)

Frame tube 3×3 cut 14.25″

Stock: Square steel tube 3 × 3 × 11 ga (0.120″) wall, A500





section, wall 0.1196″

Cut length 14.250″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

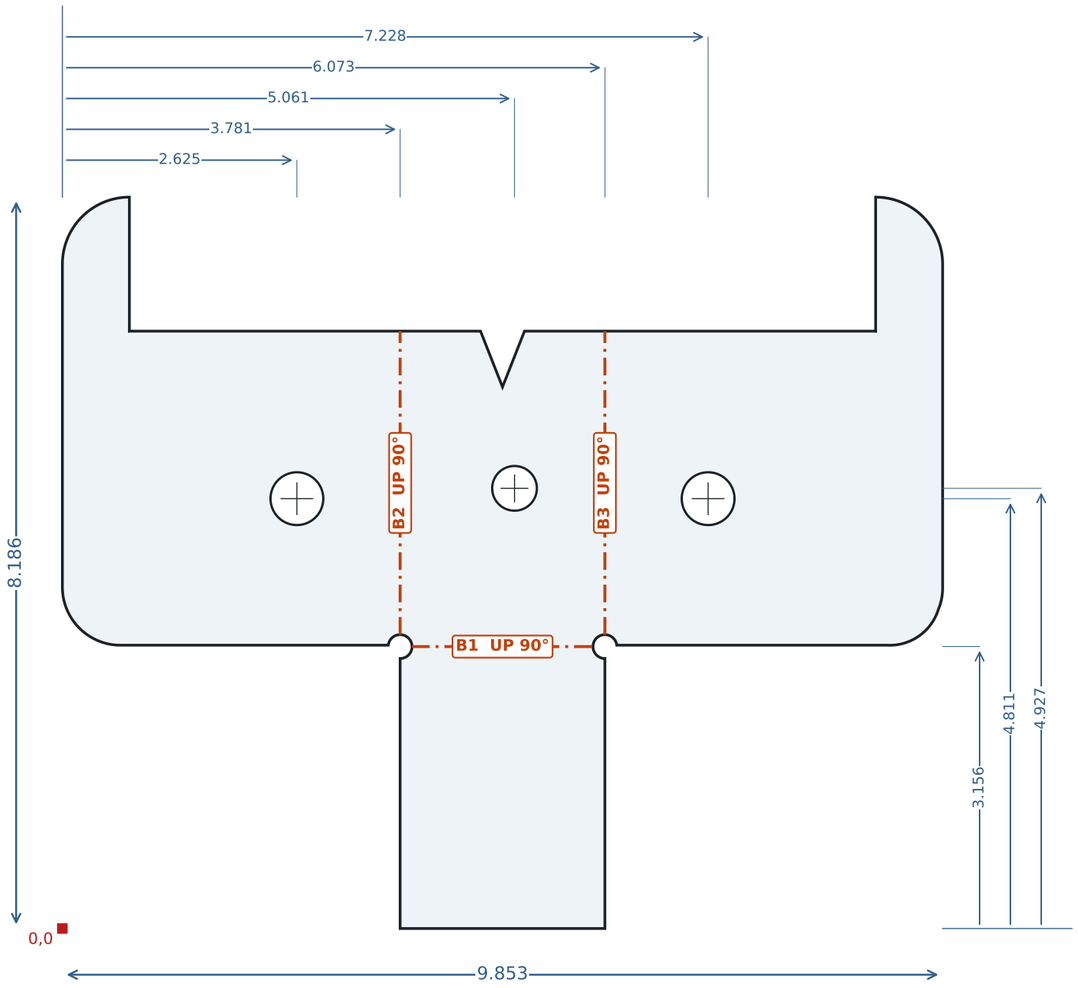
FR-6 Frame tube 3×3	**Square steel tube A500, 3 × 3**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list	**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 1/1**

FR-7 · FR-7 one piece (back + bottom + 2 sides)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 9.853″ × 8.186″ · ≈ 1.45 lb each · QTY 1 FR + 1 FL = 2 per vehicle

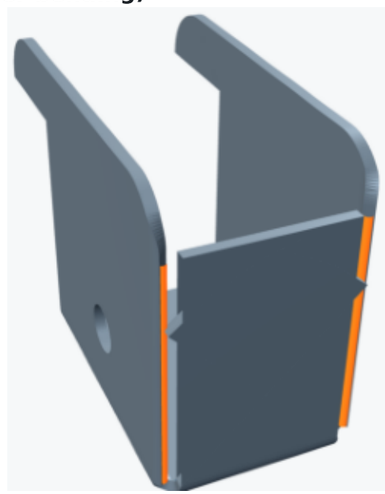
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-7 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	Y = 3.156
B2	90.0°	UP (toward)	R=t 0.120	2	X = 3.781
B3	90.0°	UP (toward)	R=t 0.120	3	X = 6.073

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	2.625	4.811	laser
H2	Ø0.500	5.061	4.927	laser
H3	Ø0.591	7.228	4.811	laser

FILES

- FR-7_FR_cut.dxf (right, qty 1 / vehicle)
- FR-7_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-7_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* one blank, 3 bends 90° the same way (R = t): bottom first, then the sides; Ø0.3 round reliefs in the two corners; after bending weld the side-to-bottom seams; V notch and Ø0.5 hole in the back, Ø0.591 holes in the sides

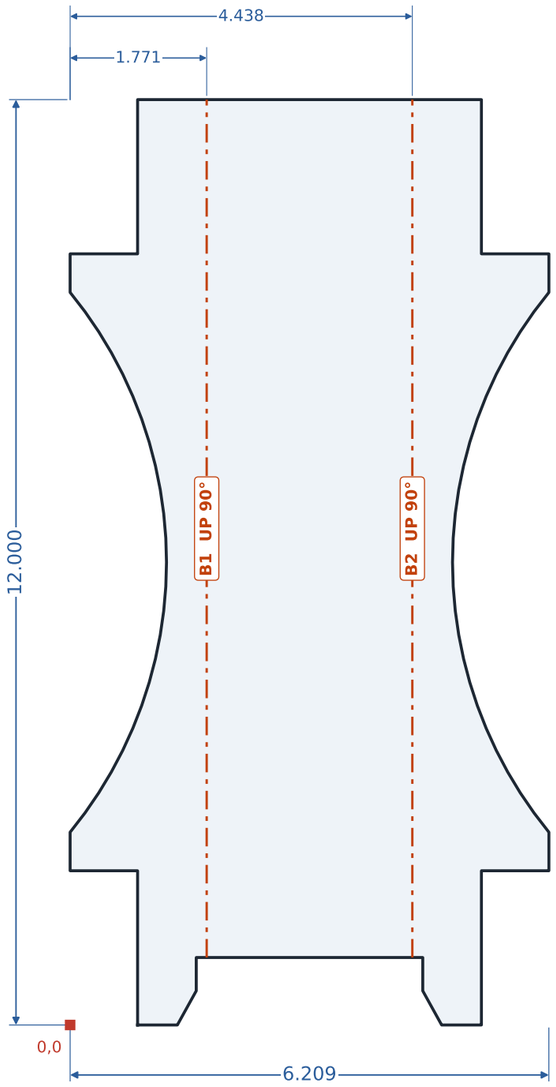
FR-7 Tray bracket (back + bottom + 2 sides)	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-7 · FR-7 one piece (back + bottom + 2 sides)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**

FR-8 · Channel (top + 2 sides)

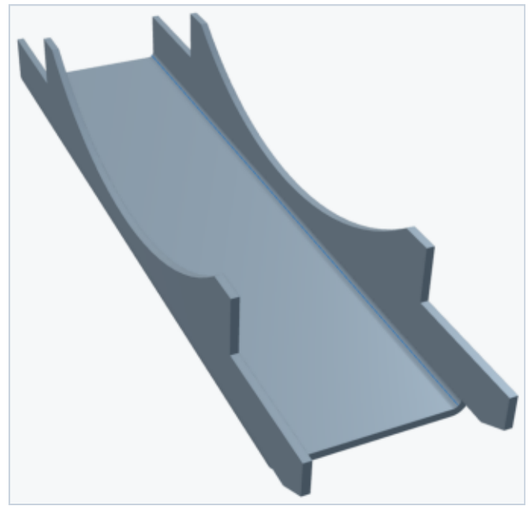
11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 6.209″ × 12.000″ · ≈ 1.79 lb each · QTY 1 FR + 1 FL = 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	X = 1.771
B2	90.0°	UP (toward)	R=t 0.120	2	X = 4.438

FILES

- FR-8_FR_cut.dxf (right, qty 1 / vehicle)
- FR-8_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-8_ref.dxf (reference: + bend lines, drill-later)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* 90° bends, R = t; relief cut at B

FR-8 Channel (top + 2 sides)	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FR-8 · Channel (top + 2 sides)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/1**

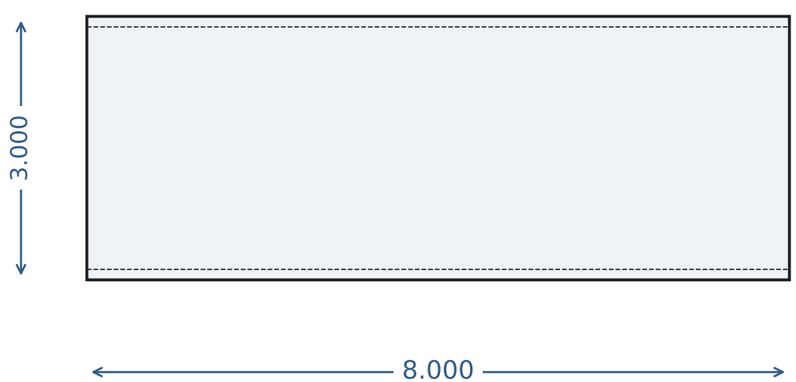
FR-9 · tube cut list

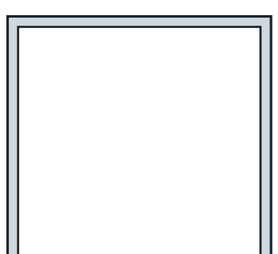
MATERIAL: Square steel tube A500, 3 × 3

WALL THICKNESS: 11 ga 0.120″ (3.04 mm)

Frame tube 3×3 cut 8″

Stock: Square steel tube 3 × 3 × 11 ga (0.120″) wall, A500





section, wall 0.1196″

Cut length 8.000″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

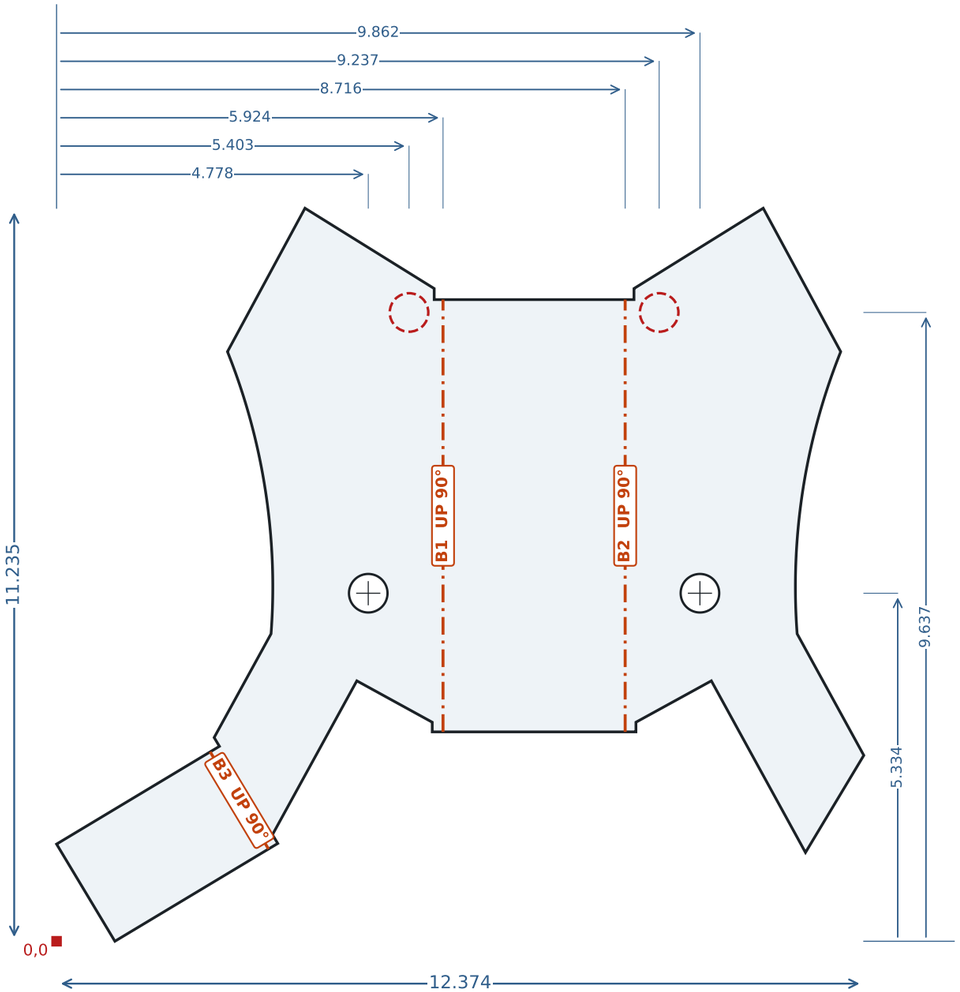
FR-9 Frame tube 3×3	**Square steel tube A500, 3 × 3**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list	**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 1/1**

FR-10-1 · Part 1 "U-body" 2 sides + wall A-H + flange E-F (3 bends)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 12.374″ × 11.235″ · ≈ 2.29 lb each · QTY 1 FR + 1 FL = 2 per vehicle

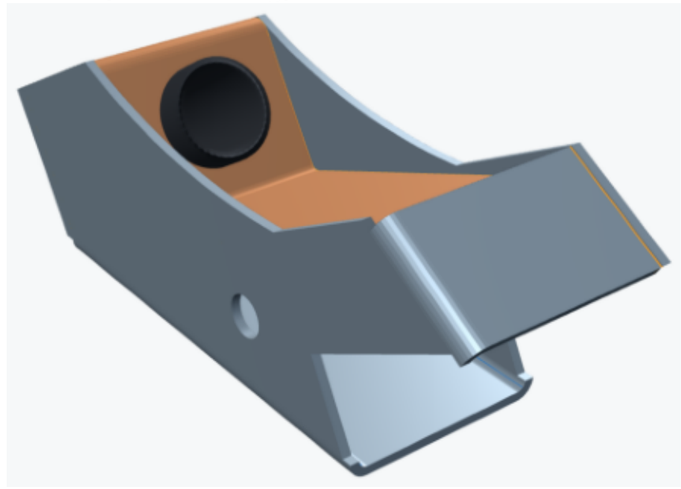
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-10-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	2	X = 5.924
B2	90.0°	UP (toward)	R=t 0.120	3	X = 8.716
B3	90.0°	UP (toward)	R=t 0.120	1	(2.354, 2.903) (3.249, 1.413)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.591	4.778	5.334	laser
H2	Ø0.591	9.862	5.334	laser
D1	Ø0.591	5.403	9.637	DRILL LATER
D2	Ø0.591	9.237	9.637	DRILL LATER

FILES

- FR-10-1_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-1_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-1_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* bends 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B) are not in the flat - drill after bending (DRILL LATER)

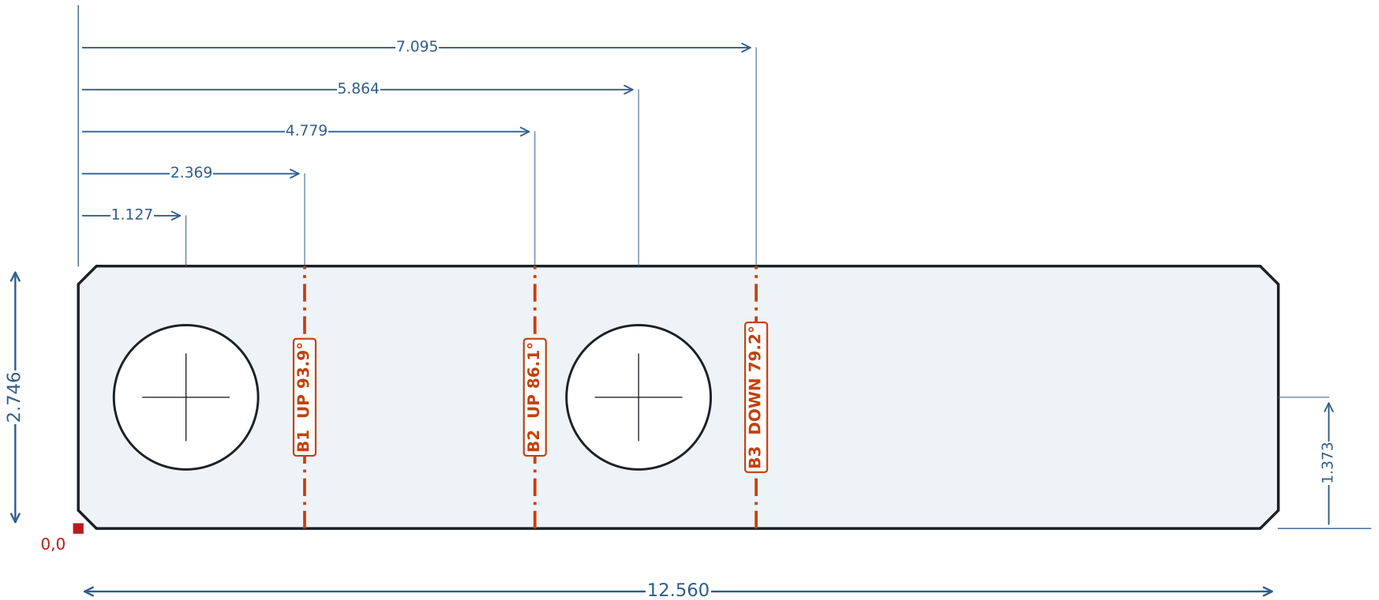
FR-10 Bracket: U-body + strip + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-10-1 · Part 1 "U-body" 2 sides + wall A-H + flange E-F (3 bends)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/3**

FR-10-2 · Part 2 "strip" A-B · B-C · shelf · inner plate (3 bends)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 12.560″ × 2.746″ · ≈ 1.05 lb each · QTY 1 FR + 1 FL = 2 per vehicle

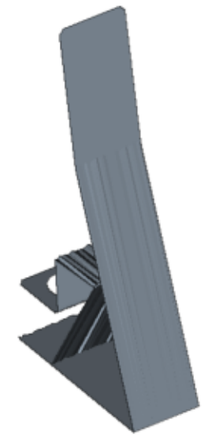
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FR-10-2 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	93.9°	UP (toward)	R=t 0.120	5	X = 2.369
B2	86.1°	UP (toward)	R=t 0.120	6	X = 4.779
B3	79.2°	DOWN (away)	R=t 0.120	4	X = 7.095

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø1.510	1.127	1.373	laser
H2	Ø1.510	5.864	1.373	laser

FILES

- FR-10-2_FR_cut.dxf (right, qty 1 / vehicle)
- FR-10-2_FL_cut.dxf (left = mirror, qty 1 / vehicle)
- FR-10-2_ref.dxf (reference: + bend lines, drill-later, material notes)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* X 79.2° opposite way; B 93.9°, C 86.1° up; corners 3/16"×45°; strip width = 3" - 2t - 0.015 (11 ga), from FR10_Bent model

FR-10 Bracket: U-body + strip + tube	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FR-10-2 · Part 2 "strip" A-B · B-C · shelf · inner plate (3 bends)	**QTY 1 FR + 1 FL = 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 2/3**

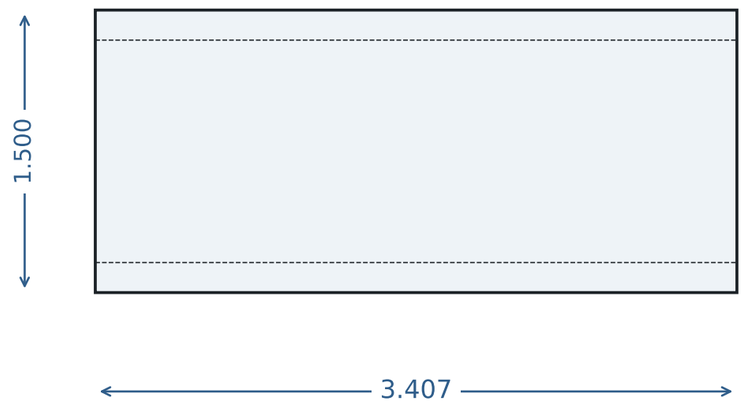
FR-10 · tube cut list

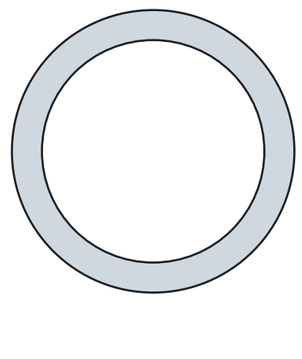
MATERIAL: DOM round steel tube A513

WALL THICKNESS: OD 1.500 × ID 1.181 (wall 0.160″)

Tube OD 1.5 / ID 1.181 cut 3.407″ (through A-B and the shelf)

Stock: DOM round steel tube OD 1.500 × ID 1.181 (wall 0.160″)





section, wall 0.1595″

Cut length 3.407″ ±1/32″, square ends, deburr

QTY 1 per side × 2 (FR + FL) = 2 per vehicle

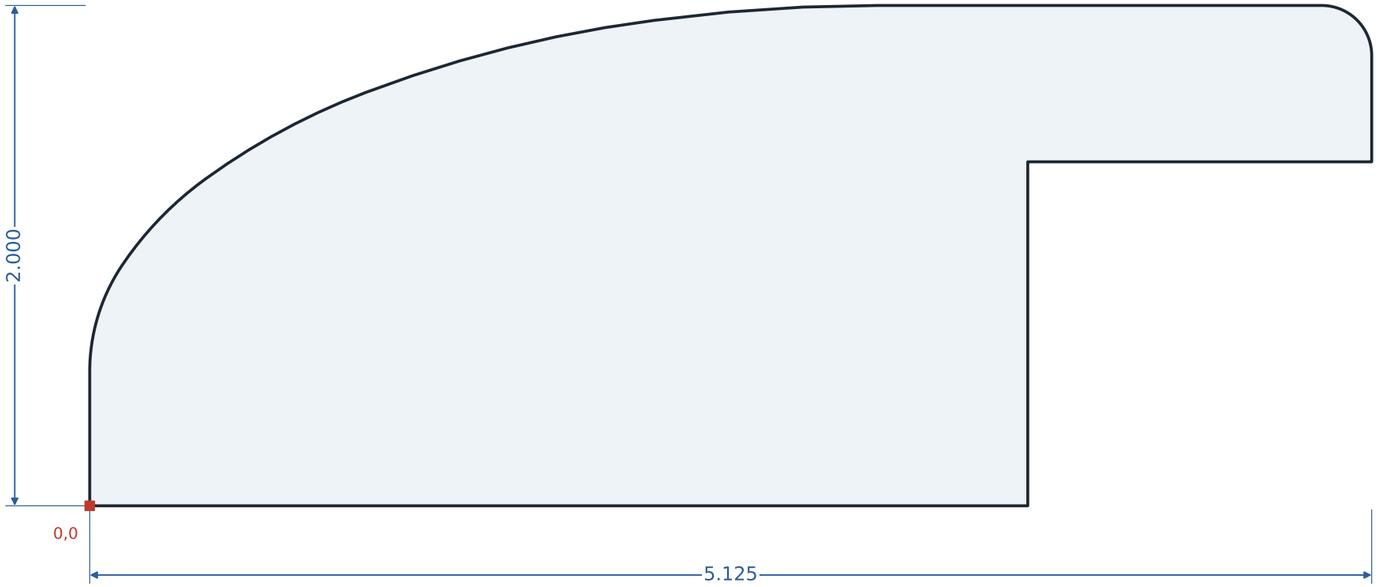
FR-10 Bracket: U-body + strip + tube	**DOM round steel tube A513**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list	**QTY 1 per side × 2 (FR + FL) = 2 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 3/3**

FR-11 · FR-11 plate (flat)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 5.125″ × 2.000″ · ≈ 0.25 lb each · QTY 2 FR + 2 FL = 4 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FILES

- FR-11_FR_cut.dxf (right, qty 2 / vehicle)
- FR-11_FL_cut.dxf (left = mirror, qty 2 / vehicle)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.
5. Right part (FR) shown. Left part (FL) = mirror: cut *_FL_cut.dxf, bend the same UP/DOWN.

* flat, no bends; 2 per side, side by side; FL-11 = mirror

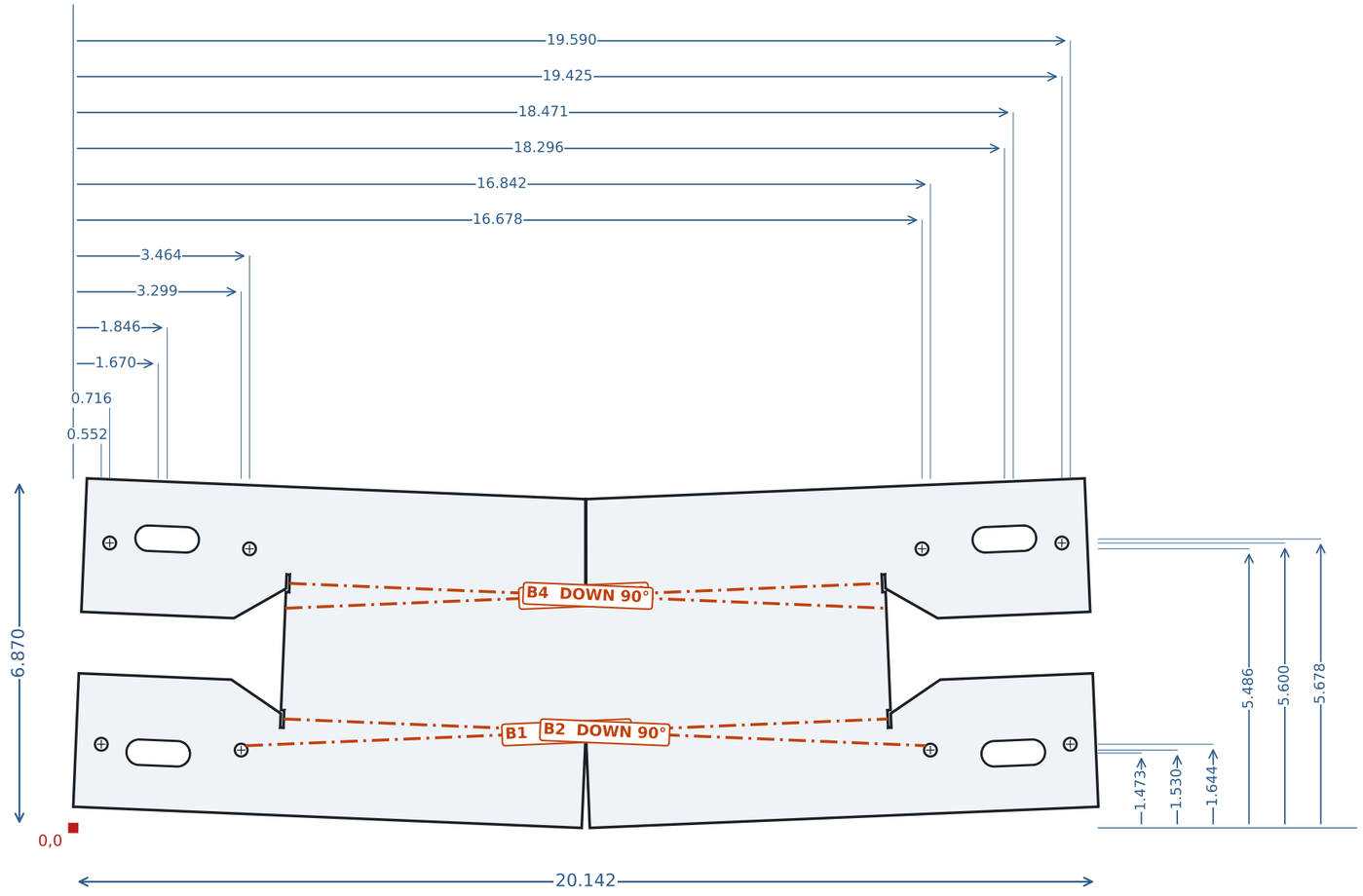
FR-11 Flat plate	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FR-11 · FR-11 plate (flat)	**QTY 2 FR + 2 FL = 4 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/1**

FF-1 · Channel with ears flat pattern, one piece (4 bends 90°, web on top)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 20.142″ × 6.870″ · ≈ 3.89 lb each · QTY 1 per vehicle

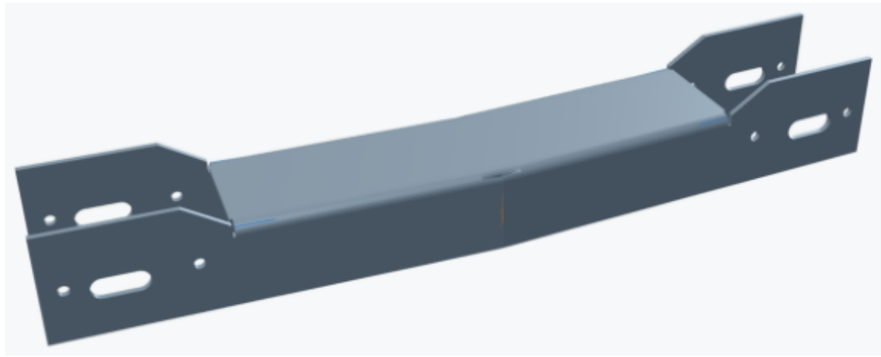
MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FF-1 · 11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS

FORMED PART (after bending)



(shape; thickness now 11 ga 0.120″)

BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	1	(3.389, 1.616) (16.004, 2.142)
B2	90.0°	DOWN (away)	R=t 0.120	2	(4.137, 2.142) (16.752, 1.616)
B3	90.0°	DOWN (away)	R=t 0.120	3	(4.166, 4.318) (15.893, 4.806)
B4	90.0°	DOWN (away)	R=t 0.120	4	(4.248, 4.806) (15.976, 4.318)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	Ø0.250	0.552	1.644	laser
H2	Ø0.250	0.716	5.600	laser
H3	Ø0.250	3.299	1.530	laser
H4	Ø0.250	3.464	5.486	laser
H5	Ø0.250	16.678	5.486	laser
H6	Ø0.250	16.842	1.530	laser
H7	Ø0.250	19.425	5.600	laser
H8	Ø0.250	19.590	1.644	laser
H9	cut-out 1.249×0.531	1.670	1.473	laser
H10	cut-out 1.249×0.531	1.846	5.678	laser
H11	cut-out 1.249×0.531	18.296	5.678	laser
H12	cut-out 1.249×0.531	18.471	1.473	laser

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. MATERIAL HR steel sheet A1011 CS, 11 ga t = 0.1196″ (3.04 mm). Bend allowance (K = 0.44, inside radius = t) is already in the flat.
   Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut - drill after bending/welding.
4. Deburr all edges, remove dross.

* one blank: the web is a chevron 6"+6" (ends ¼" out); bend the walls 90° down on the dashed lines, each in 2 hits (6" to the middle); outer wall has a V-notch - closes after bending, weld it; inner wall has a slit - opens after bending, weld it; the ears reach past the bend line; 1/16" relief slit at the bend end

FILES

- FF-1_cut.dxf (qty 1 / vehicle)
- FF-1_ref.dxf (reference: + bend lines, material notes)

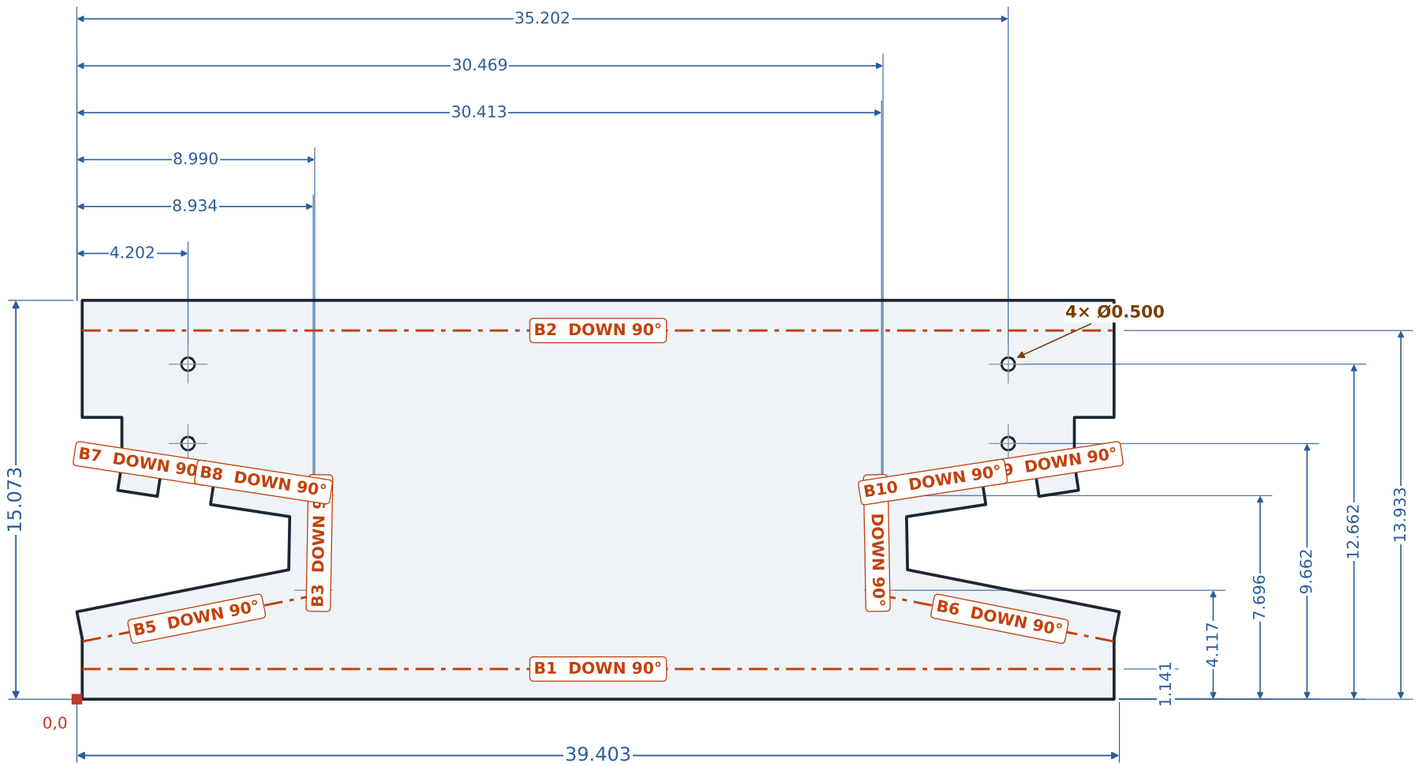
FF-1 Front channel with ears (chevron 6"+6")	**HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
FF-1 · Channel with ears flat pattern, one piece (4 bends 90°, web on top)	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-31	**SHEET 1/1**

FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down

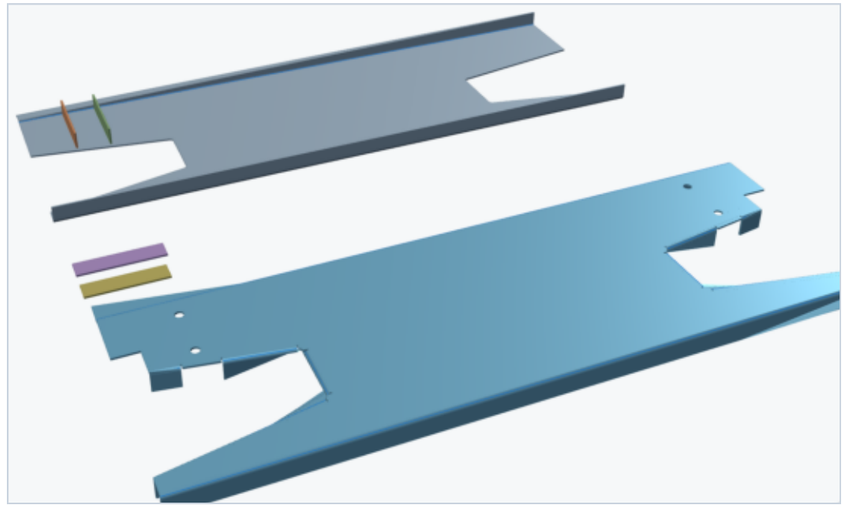
11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.403″ × 15.073″ · ≈ 17.70 lb each · QTY 1 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196″ (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FORMED PART (after bending)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	DOWN (away)	R=t 0.120	3	Y = 1.141
B2	90.0°	DOWN (away)	R=t 0.120	4	Y = 13.933
B3	90.0°	DOWN (away)	R=t 0.120	5	(9.138, 4.256)→(9.197, 7.546)
B4	90.0°	DOWN (away)	R=t 0.120	6	(30.265, 4.256)→(30.206, 7.546)
B5	90.0°	DOWN (away)	R=t 0.120	7	(0.222, 2.185)→(8.833, 3.892)
B6	90.0°	DOWN (away)	R=t 0.120	8	(39.181, 2.185)→(30.570, 3.892)
B7	90.0°	DOWN (away)	R=t 0.120	9	(1.717, 9.022)→(3.212, 8.793)
B8	90.0°	DOWN (away)	R=t 0.120	10	(5.223, 8.486)→(8.878, 7.926)
B9	90.0°	DOWN (away)	R=t 0.120	11	(37.686, 9.022)→(36.191, 8.793)
B10	90.0°	DOWN (away)	R=t 0.120	12	(34.180, 8.486)→(30.526, 7.926)

HOLES (centre from 0,0)

Hole	Size	X	Y	Made by
H1	cut-out 0.513×0.516	30.326	7.802	laser
H2	Ø0.500	35.202	12.662	laser
H3	Ø0.500	35.202	9.662	laser
H4	cut-out 0.514×0.504	30.387	4.011	laser
H5	cut-out 0.513×0.516	9.077	7.802	laser
H6	cut-out 0.514×0.504	9.016	4.011	laser
H7	Ø0.500	4.202	12.662	laser
H8	Ø0.500	4.202	9.662	laser

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* rims 1⅛" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FILES

- FF-3-1_cut.dxf (qty 1 / vehicle)
- FF-3-1_ref.dxf (reference: + bend lines, drill-later)

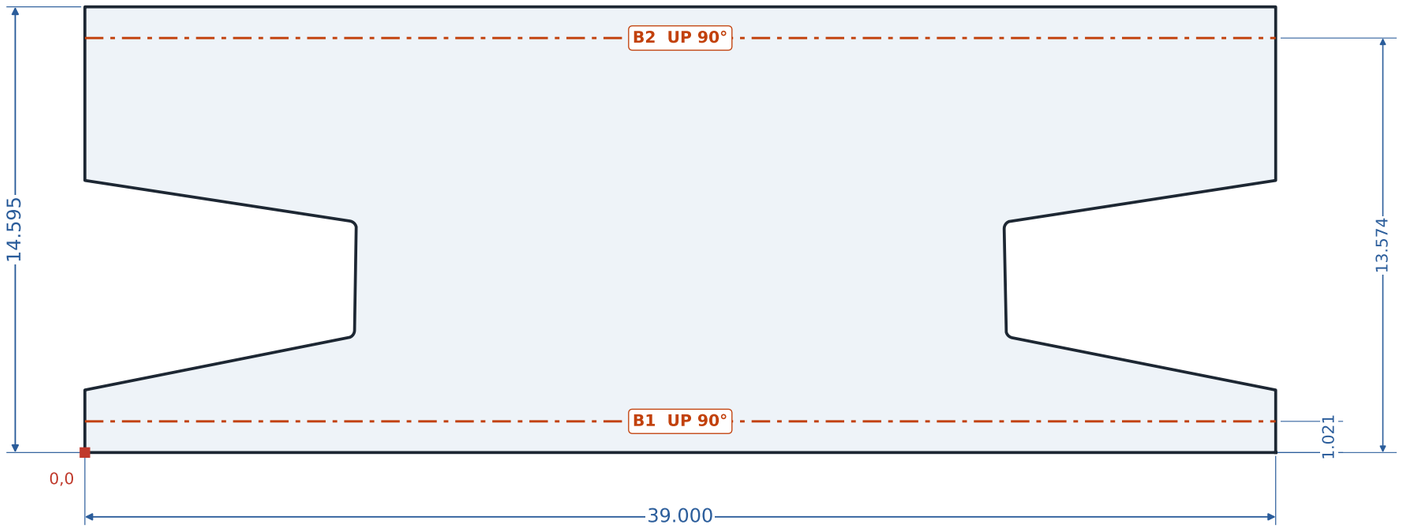
FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-1 · FF3-1 top plate — flat pattern, 10 bends 90° down	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 1/4**

FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 39.000″ × 14.595″ · ≈ 16.12 lb each · QTY 1 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



BEND TABLE

Bend	Angle	Direction	Inside R	Order	Bend line position
B1	90.0°	UP (toward)	R=t 0.120	1	Y = 1.021
B2	90.0°	UP (toward)	R=t 0.120	2	Y = 13.574

FILES

- FF-3-2_cut.dxf (qty 1 / vehicle)
- FF-3-2_ref.dxf (reference: + bend lines, drill-later)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* rims on the long sides bent up 1" (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16" reliefs at the bend ends

FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-2 · FF3-2 lower plate — flat pattern, 2 bends 90° up	**QTY 1 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 2/4**

FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 4.687″ × 1.125″ · ≈ 0.18 lb each · QTY 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FILES
- FF-3-3_cut.dxf (qty 2 / vehicle)

FLAT PATTERN (as laser-cut, marked side up)



NOTES
1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

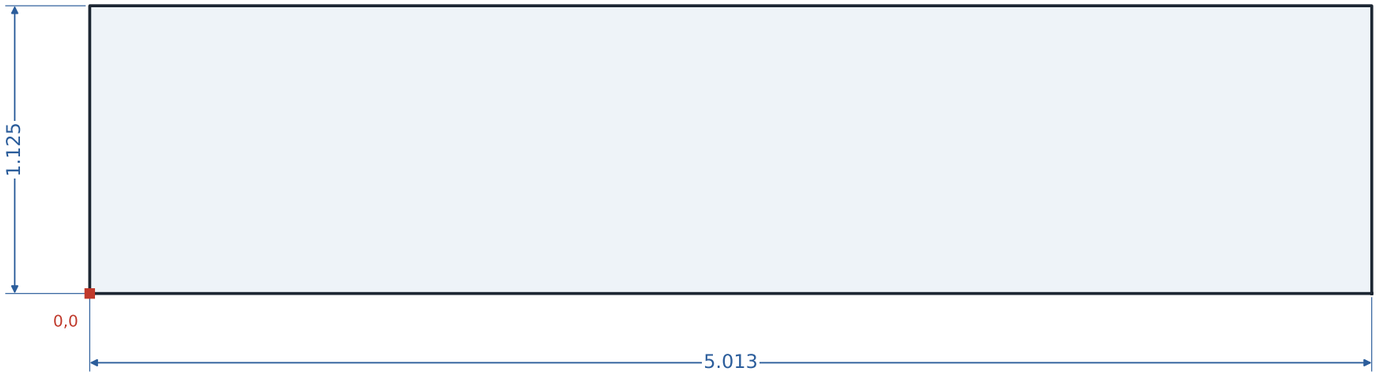
FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-3 · Passage wall 1⅛ × 4.69" (instead of the tube)	**QTY 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 3/4**

FF-3-4 · Passage wall 1⅛ × 5.01" (instead of the tube)

11 ga 0.120″ (3.04 mm) · HR steel sheet A1011 CS · flat blank 5.013″ × 1.125″ · ≈ 0.19 lb each · QTY 2 per vehicle

MATERIAL: HR steel sheet A1011 CS
THICKNESS: 11 ga t = 0.1196'' (3.04 mm)

FLAT PATTERN (as laser-cut, marked side up)



FILES

- FF-3-4_cut.dxf (qty 2 / vehicle)

NOTES

1. Laser cut from the *_cut.dxf file (1:1, inch). The DXF governs the geometry; dimensions here are for checking.
2. Bend allowance (K = 0.44, inside radius = t) is already in the flat. Bend on the orange lines; UP = toward you when the part lies as drawn.
3. Holes marked DRILL LATER (red dashed) are not laser cut — drill after bending/welding.
4. Deburr all edges, remove dross.

* stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 Cross box: top plate + lower plate + 4 passage walls	HR steel sheet A1011 CS 11 ga 0.120″ (3.04 mm)	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV A**
FF-3-4 · Passage wall 1⅛ × 5.01" (instead of the tube)	**QTY 2 per vehicle**	Tol.: cut ±0.010″ · bend ±1° · IR = t	2026-10-03	**SHEET 4/4**

FF-4 · tube cut list

MATERIAL: Square steel tube A500, 1 ½ × 1 ½

WALL THICKNESS: 11 ga 0.120″ (3.04 mm)

Square tube 1 ½ cut 39″

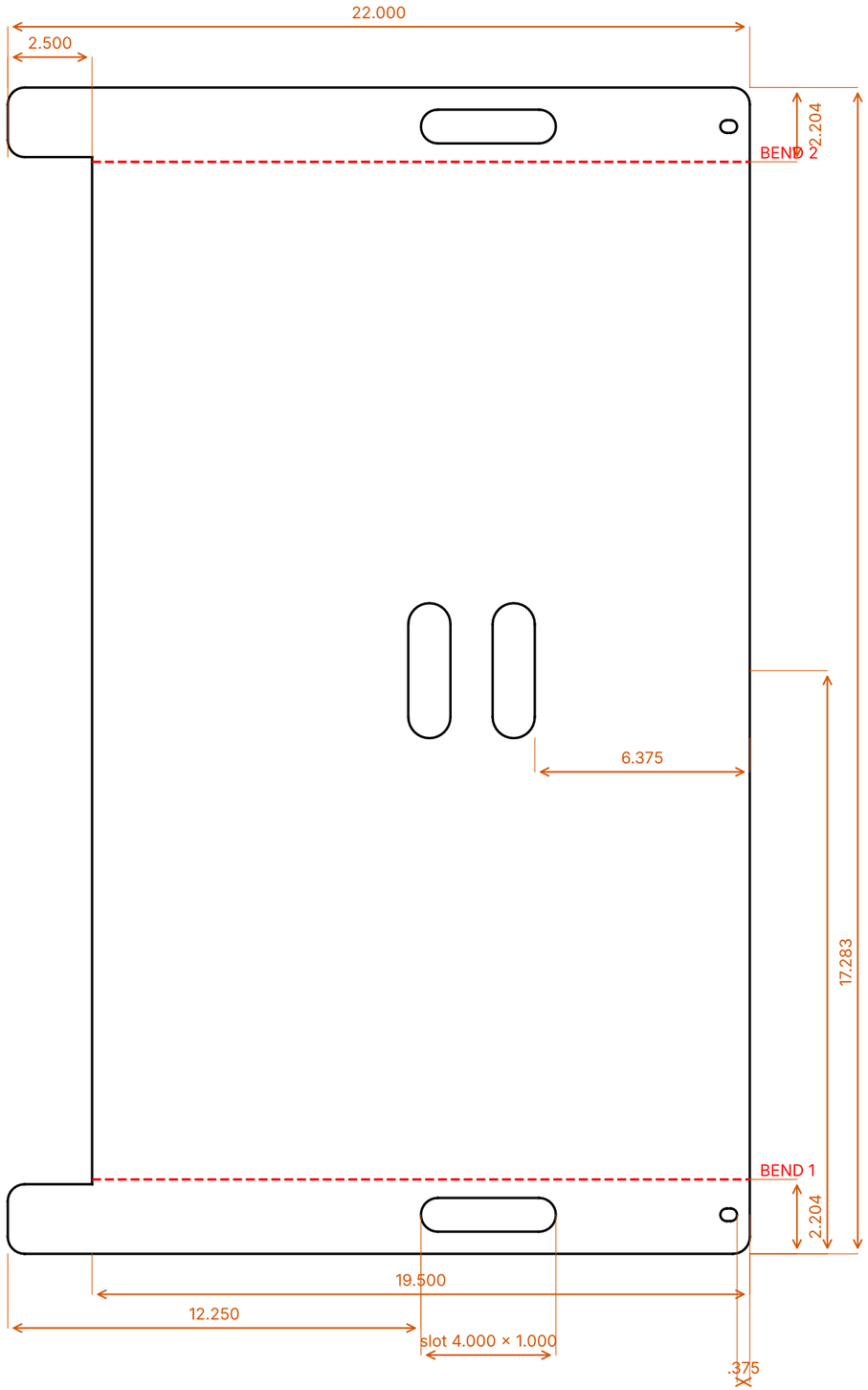
Stock: Square steel tube 1 ½ × 1 ½ × 11 ga (0.120″) wall, A500



Cut length 39.000″ ±1/32″, square ends, deburr

QTY 1 per vehicle

FF-4 Square tube 1 ½	**Square steel tube A500, 1 ½ × 1 ½**	Units: inch · NOT TO SCALE	Sienna WAV frame	**REV B**
Tube cut list	**QTY 1 per vehicle**	Tol.: cut ±1/32″	2026-10-31	**SHEET 1/1**



RA-1 RAMP PLATE (Sienna WAV)

Qty: 2 per 2 vehicles (1 per vehicle)

Material: Aluminum 5052-H32, 0.125" (1/8")

Units: inch. Cut tolerance ±0.010", bend ±1°

Flat size: 22.000 × 34.565

BEND TABLE

#	line Y	angle	direction	inside R
1	2.204	90°	UP (same side)	0.125
2	32.362	90°	UP (same side)	0.125

K = 0.44, BA = 0.2827, BD = 0.2173

Bend order: 1 then 2. Bend lines from X = 0 to 19.500

FINISHED PART (outside dims)

Plate 19 1/2 × 30 3/8 (outside of the ears)

Ears 22 long x 2 5/16 high, 90°, same side (tray)

Ear right end flush with plate edge; 2 1/2 overhang on the left

Ear ends R 1/2

HOLES / SLOTS

Ears: handle slot 4 × 1 at 12 1/4 from ear left end, centred on ear height

Ears: slot 3/8 × 1/2 at 3/8 from ear right end (edge), centred on ear height

Plate: 2 slots 1 1/4 × 4, gap 1 1/4, 6 3/8 from long edge, centred

POINTS (finished part)

A-B 22 B-I 2 5/16 I-J 30 3/8 J-C 2 5/16 C-D 22

F-G 30 3/8 (plate edge) plate width 19 1/2

Files: RA-1_cut.dxf (CUT only), RA-1_ref.dxf (+ BEND dashed)

Model: RA1.FCStd (finished), RA1_flat.FCStd (flat)

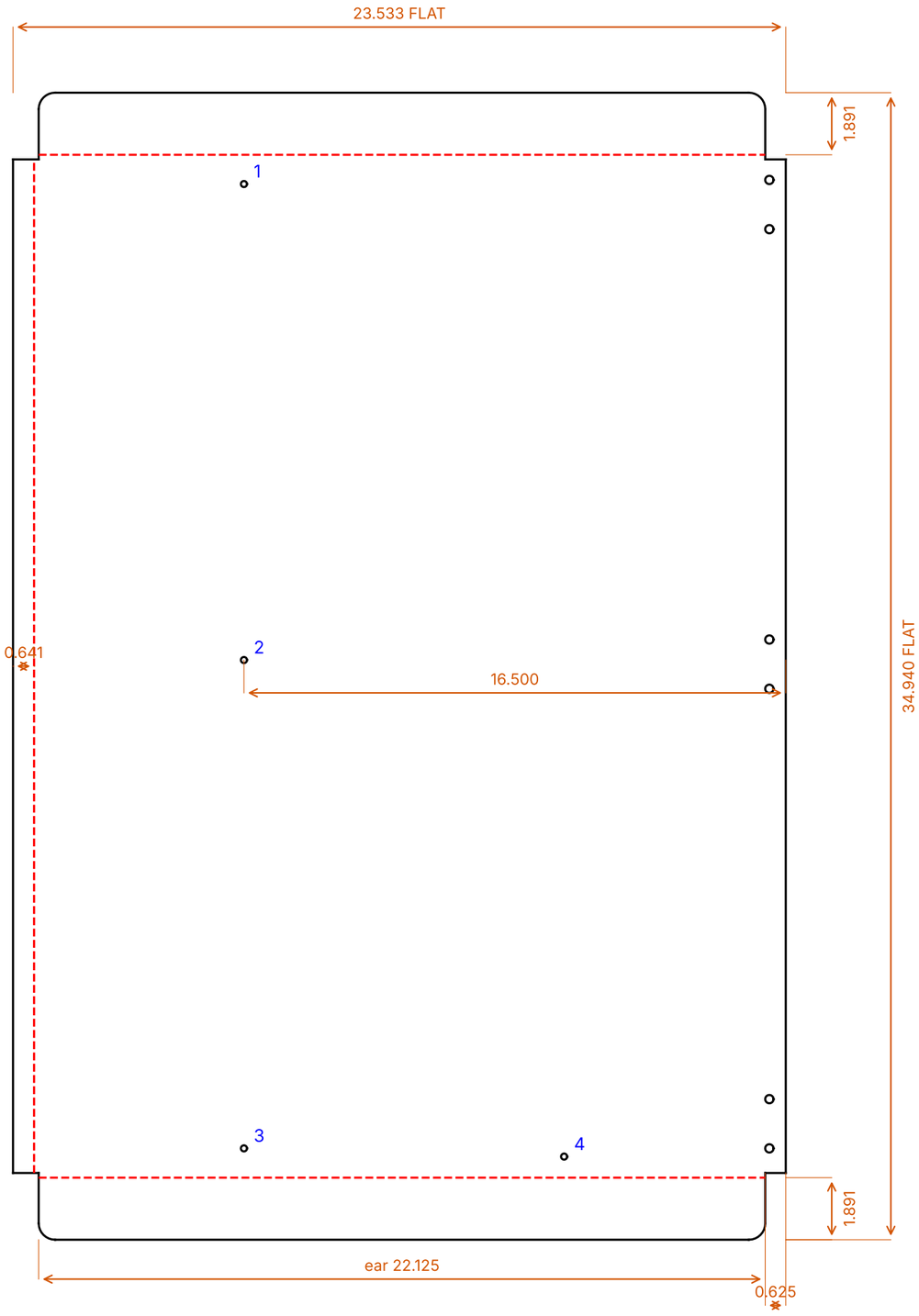
Rev A 2026-10-27



END VIEW 30 3/8 outside, ears 2 5/16, IR 1/8

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)



RA-2 RAMP PLATE (Sienna WAV)

Qty: 2 per 2 vehicles (1 per vehicle)

Material: Aluminum 5052-H32, 0.125" (1/8")

Units: inch. Cut ±0.010", bend ±1°

Flat size: 23.533 × 34.940

BEND TABLE (all 90°, DOWN = away from the viewer, as drawn)

1 flange 3/4 line X = 0.641 90° DOWN IR 0.125

2 ear bottom line Y = 1.891 90° DOWN IR 0.125

3 ear top line Y = 33.049 90° DOWN IR 0.125

K 0.44, BA 0.2827, BD 0.2173. Bend order: 1, 2, 3

FINISHED PART (outside dims)

Plate 23 (flange outer face to right edge) x 31 3/8 (ear outer faces)

Flange 3/4 high along the left side

Ears 2 high, from the flange side, stop 5/8 short of the right edge

Ear ends R 1/2. Corner reliefs where flange meets ears.

HOLES

1, 2, 3 (Ø 3/16): 16 1/2 from right edge; 1 from top, 15 7/8 and 1 from bottom

4: 3/4 from bottom, ~6 3/4 from right edge (PRELIMINARY)

Right edge: 6 x Ø6.5 mm, centre 1/2 from edge,

at 1, 2 1/2, 15, 16 1/2, 29, 30 1/2 (from the lower ear side)

POINTS (finished part)

A-H flange 30.875 B-C, G-F ears 22.125 D-E 31 3/8

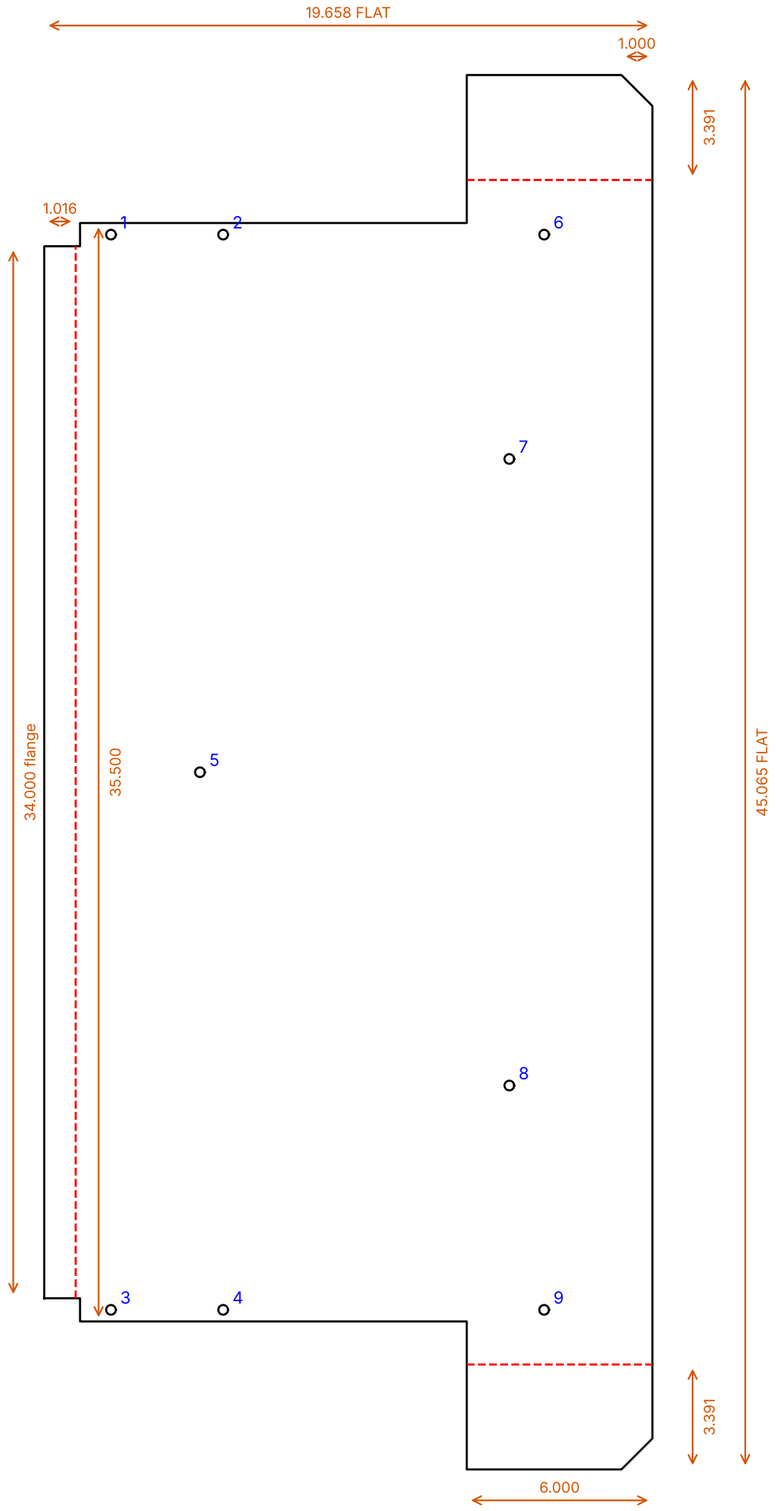
Files: RA-2_cut.dxf (CUT), RA-2_ref.dxf (+ BEND dashed)

Model: RA2.FCStd (finished), RA2_flat.FCStd (flat)

Rev B 2026-10-31

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)



RA-3 RAMP PLATE (Sienna WAV)

Qty: 1 per vehicle

Aluminum 5052-H32 0.125" (1/8")

Units inch. Cut ±0.010", bend ±1°

Flat 19.658 × 45.065

BEND TABLE (all 90°, DOWN as drawn, IR 0.125)

1 flange line X = 1.016 (Y 5.533..39.533)

2 ear low line Y = 3.391 (X 13.658..19.658)

3 ear top line Y = 41.674

K 0.44, BA 0.2827, BD 0.2173

FINISHED PART (outside dims)

Flange 1 1/8 along the left, 34 long, centred

Plate 12 3/4 + strip 6 (flat 19 7/8 with flange)

Ears on the strip: bend lines 38 1/2 apart, ears 3 1/2, 6 → 5 at tip

Plate ends 1 1/2 before each ear bend

HOLES Ø 8 mm (flat coordinates from lower-left corner)

1 X 2.158 Y 39.908

2 X 5.783 Y 39.908

3 X 2.158 Y 5.158

4 X 5.783 Y 5.158

5 X 5.033 Y 22.533

6 X 16.158 Y 39.908

7 X 15.033 Y 32.658

8 X 15.033 Y 12.408

9 X 16.158 Y 5.158

1,2 / 3,4: 1 1/4 & 4 7/8 from flange bend, 3/8 from plate edge

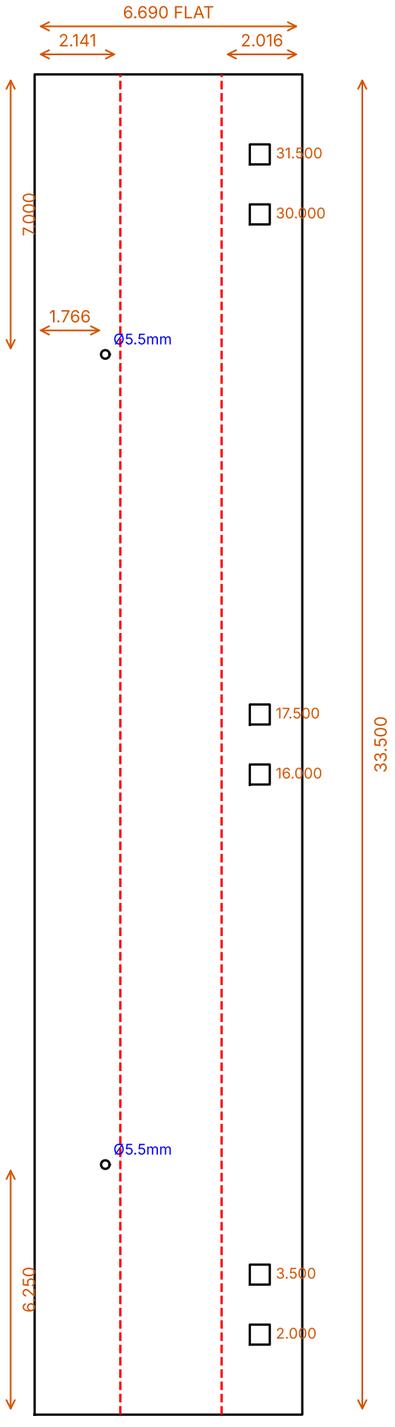
5: 14 5/8 from right, 17 3/4 from top plate edge

6,9: 3 1/2 from right, 1 7/8 from ear bend; 7,8: 4 5/8 from right, 9 1/8 from ear bend

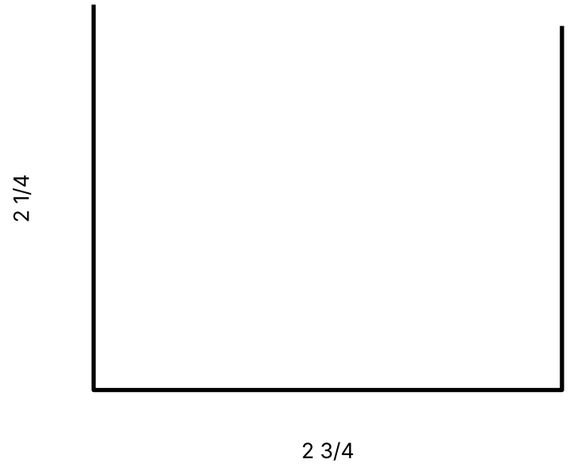
Files: RA-3_cut.dxf, RA-3_ref.dxf Model: RA3.FCStd, RA3_flat.FCStd Rev A 2026-10-31

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)



SECTION (outside dims), IR 1/8



RA-4 RAMP CHANNEL (Sienna WAV)

Qty: 1 per vehicle

Aluminum 5052-H32 0.125" (1/8")

Units inch. Cut ±0.010", bend ±1°

Flat 6.690 × 33.500

BEND TABLE (90°, DOWN as drawn, IR 0.125)

1 line X = 2.141 90° DOWN

2 line X = 4.674 90° DOWN

K 0.44, BA 0.2827, BD 0.2173

HOLES

Wall 2 1/4: 2 x Ø5.5 mm, 3/8 from bend line,

7 from top end, 6 1/4 from bottom end

Wall 2 1/8: 6 x square 1/2 × 1/2, centred on wall height,

centres 2 & 3 1/2 from top, 16 & 17 1/2 and 2 & 3 1/2 from bottom

POINTS: A-B 2 1/4, B-C 2 3/4, C-D 2 1/8, D-E 33 1/2

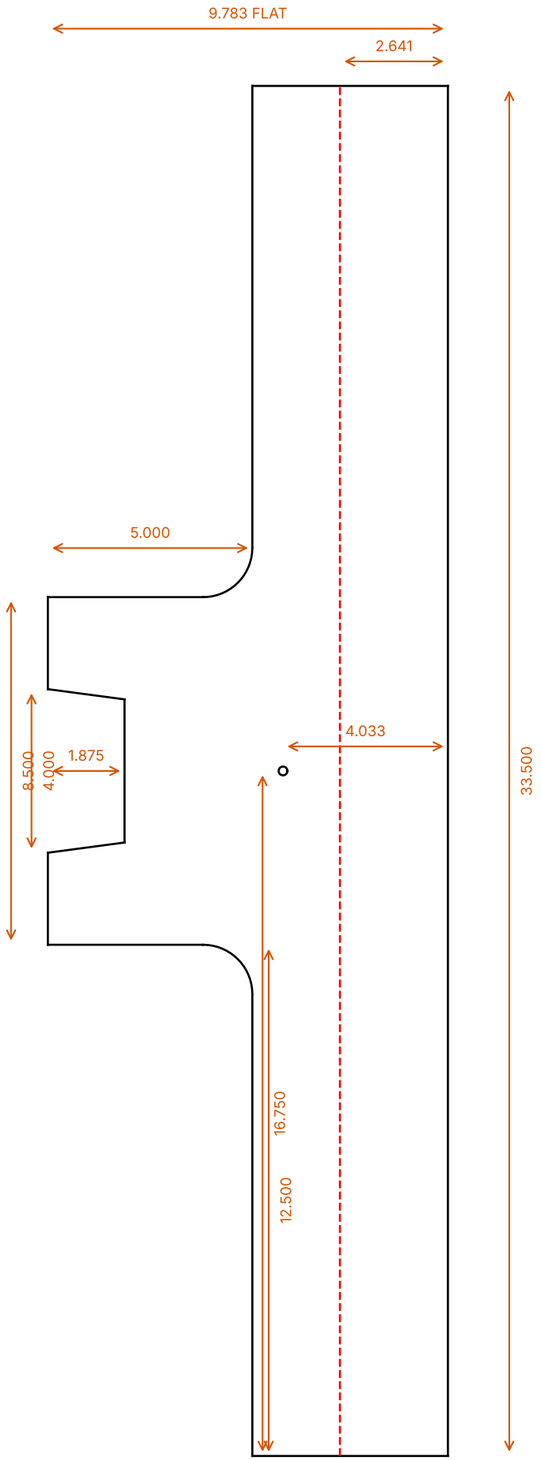
Files: RA-4_cut.dxf, RA-4_ref.dxf

Model: RA4.FCStd, RA4_flat.FCStd

Rev A 2026-10-27

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)



RA-5 RAMP ANGLE WITH TAB (Sienna WAV)

Qty: 1 per vehicle

Aluminum 5052-H32 0.125" (1/8")

Units inch. Cut ±0.010", bend ±1°

Flat 9.783 × 33.500

BEND TABLE

1 line X = 7.141   90° UP (toward the viewer)   IR 0.125

K 0.44, BA 0.2827, BD 0.2173

FINISHED PART (outside dims)

Flange 2 3/4 high along the right edge, 90° UP

Strip 5 wide (incl. flange), tab 5 out (10 overall), tab from 12 1/2 to 21

Notch in tab tip: opening 4, bottom 3 1/2, depth 1 7/8

Tab/strip fillets R 1.207 (arc mid 1/2 from the corner)

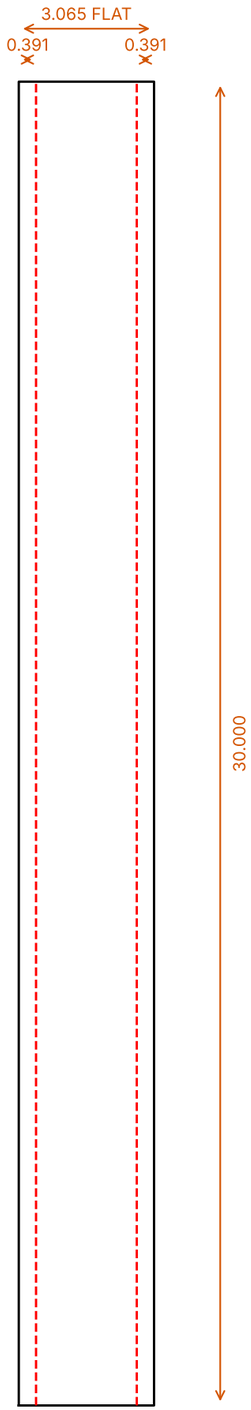
HOLE

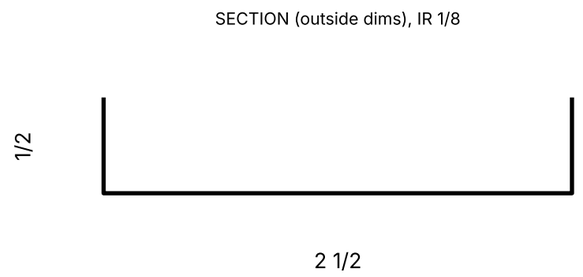
Ø 5.5 mm: 1 1/2 from the bend line (4 1/4 from right edge), 16 3/4 from bottom

Files: RA-5_cut.dxf, RA-5_ref.dxf   Model: RA5.FCStd, RA5_flat.FCStd   Rev A 2026-10-31

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)





RA-6 RAMP SPACER CHANNEL (Sienna WAV)

Qty: 1 per vehicle

Aluminum 5052-H32 0.125" (1/8")

Units inch. Cut ±0.010", bend ±1°

Flat 3.065 × 30.000

BEND TABLE (90°, DOWN as drawn, IR 0.125)

1 line X = 0.391 90° DOWN

2 line X = 2.674 90° DOWN

K 0.44, BA 0.2827, BD 0.2173

HOLES: none

POINTS: A-B 1/2, B-C 2 1/2, C-D 1/2, D-E 30

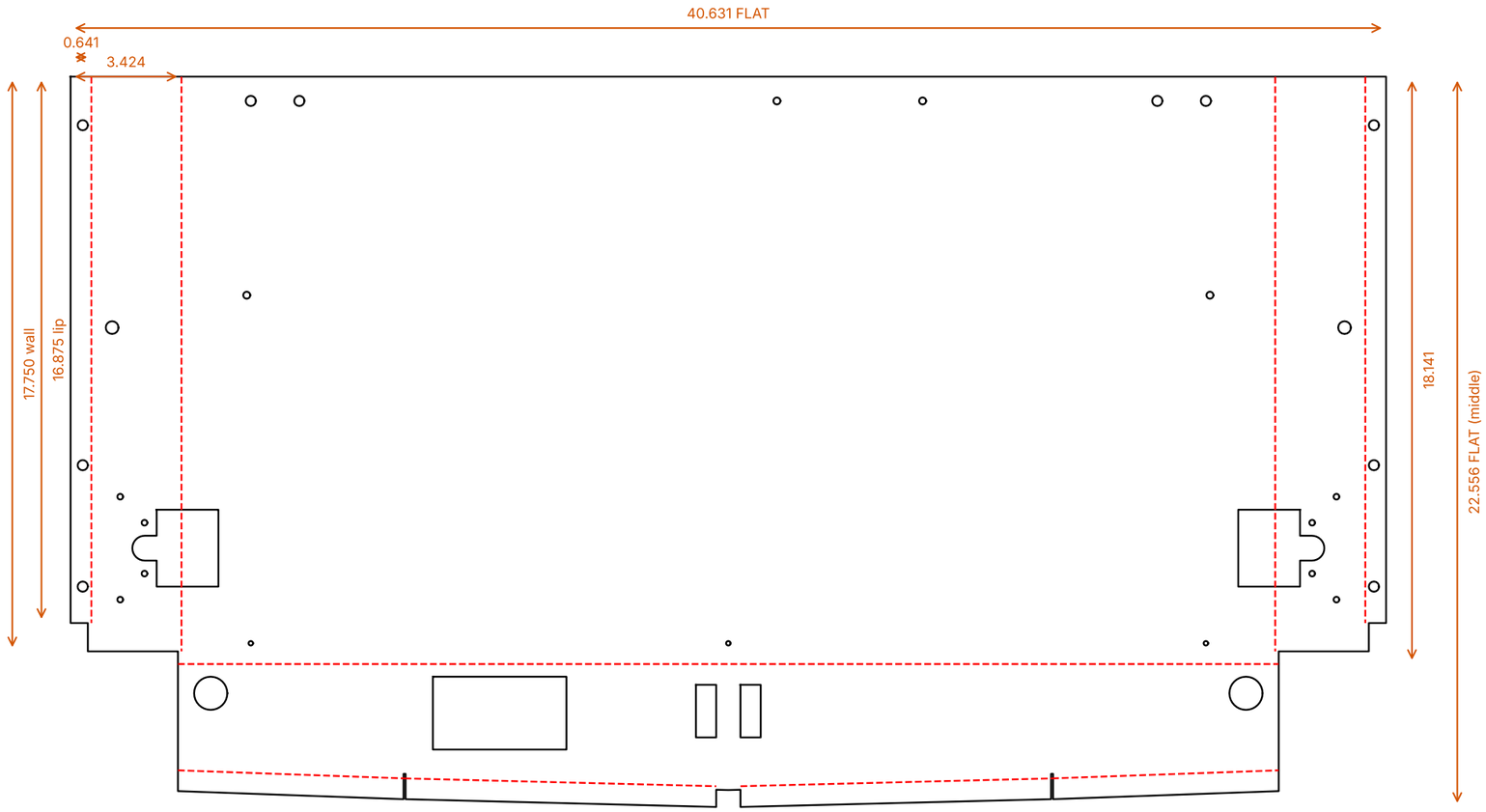
Files: RA-6_cut.dxf, RA-6_ref.dxf

Model: RA6.FCStd, RA6_flat.FCStd

Rev A 2026-10-27

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)



RA-7 RAMP BOX (Sienna WAV)

Qty: 1 per vehicle
Aluminum 5052-H32 0.125" (1/8")
Units inch. Cut ±0.010", bend ±1°
Flat 40.631 × 22.556

BENDS (IR 0.125, K 0.44, BD 0.2173)

Side walls: 90° DOWN at X 3.424 / 37.207
Side lips: 90° OUTWARD at X 0.641 / 39.990
Front wall: 90° DOWN at Y 18.141
Front lip: 4 straight segments, 90° OUTWARD,
  relief slits 1/16 at the kinks (x 7, 27)
Order: side lips, side walls, front lip, front wall

FINISHED PART (outside dims)

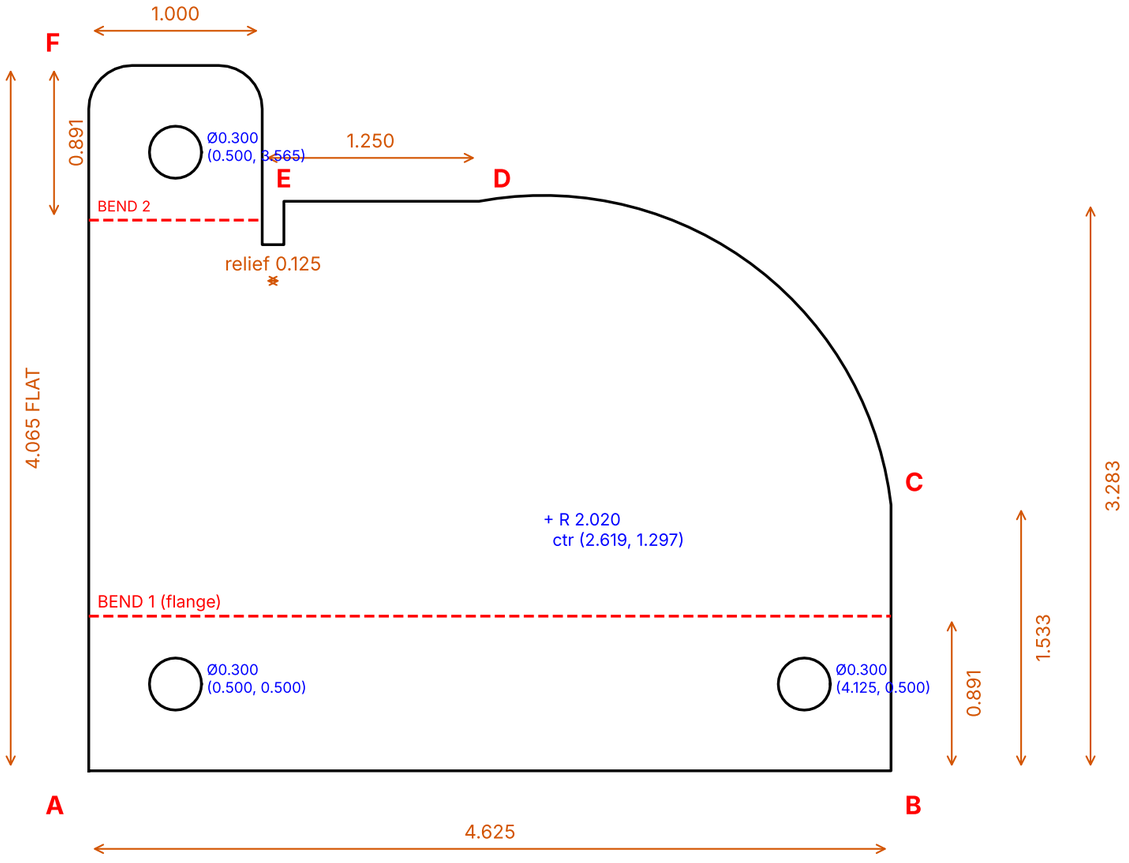
Top 34 × 18 1/4, no back wall
Side walls 3 × 17 3/4, lip 3/4 (16 7/8 long)
Front wall 3 1/2 ends / 3 3/4 at 7 / 4 middle, lip 3/4:
  2 × 16 5/8 with 3/4 gap in the middle

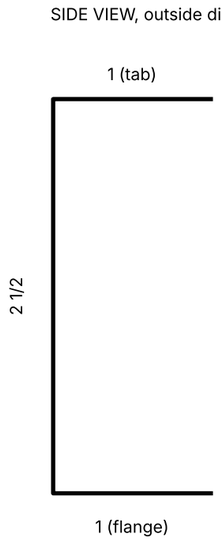
HOLES

Top: Ø8 mm 3/4 from back, 2 1/4 & 3 3/4 from sides
  Ø5.5 mm 6 3/4 from back, 2 1/8 from sides
  Ø3.5 mm 3/4 from front, 2 1/4 from sides + middle
  Ø5.5 mm 3/4 from back, 11 & 15 1/2 from one side
  corner cutouts 2 3/8 × 1 1/4 at 13 3/8 from back
Side wall: Ø10 mm 3/4 up, 7 3/4 from back; U-cutout 2 3/8,
  U Ø19.5 mm to 1 3/8, screws Ø4.5 mm (A,D 1; B,C 1 3/4)
Side lip: 3 x Ø8 mm at 1 1/2, 12, 15 3/4 from back
Front wall: Ø26 mm at both ends (edge 1/2, 1/2);
  cutout 7 7/8..12 x (1/2..2 3/4); squares 5/8 × 1 5/8,
  3/4 down, 3/4 apart, centred

Files: RA-7_cut.dxf, RA-7_ref.dxf Model: RA7.FCStd, RA7_flat.FCStd Rev A 2026-10-31

MATERIAL: Aluminum 5052-H32 sheet
THICKNESS: t = 0.125″ (1/8″, 3.18 mm)





RA-8 RAMP BRACKET (Sienna WAV)

Qty: 2 per vehicle

Aluminum 5052-H32 0.125" (1/8")

Units inch. Cut ±0.010", bend ±1°

Flat 4.625 × 4.065

FINISHED (outside): 4 5/8 × 2 1/2

flange 1 full length, tab 1 × 1 (top-left)

tab straight 3/4, corners R 1/4

arc C-D R 2.020: D 1 1/4 right of tab, C 3/4 up,
passes X 4 1/2 at Y 1 1/4

BEND TABLE (IR 0.125, both 90° SAME side)

1 Y = 0.891 full width (flange)

2 Y = 3.174 X 0..1 (tab)

K 0.44, BA 0.2827, BD 0.2173

HOLES Ø0.300 (7.6 mm) x3

flange: 1/2 from ends, centred on flange

tab: centred, 1/2 from tip

Files: RA-8_cut.dxf, RA-8_ref.dxf

Model: RA8.FCStd, RA8_flat.FCStd

Rev A 2026-10-31

MATERIAL: Aluminum 5052-H32 sheet

THICKNESS: t = 0.125″ (1/8″, 3.18 mm)